Toward Generalizable Deepfake Video Detection

by

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ABSTRACT

Deepfake video detection systems intended for forensic screening and other consequential applications must support decisions about both video authenticity and when further review is needed. Detectors often perform well on familiar manipulations but fail under new generation methods, recording conditions, or post-processing, sometimes assigning high confidence to incorrect predictions. This dissertation investigates how complementary facial representations and uncertainty-aware learning can improve generalization and help identify unreliable detection decisions. Three connected studies address this problem. The first, DBaGNet, combines identity, behavioral, and geometric signatures to learn representations that transfer across manipulation methods and datasets. The second, DISCERN, integrates visual, semantic, and structural evidence through evidential learning and disagreement-aware training, extending detection to uncertainty estimation and selective prediction. Building on these studies, the proposed CROSS-EXAM framework combines facial appearance, latent facial structure, and video dynamics within a three-branch spatio-temporal framework. It learns the views separately before jointly refining their evidential predictions and integrating them through reliability-aware conflict redistribution. A one-sided disagreement objective distinguishes conflicting evidence from shared uncertainty, addressing the risk of encouraging confident predictions when all views lack sufficient evidence. Preliminary results indicate complementary strengths of spatial and video representations, particularly for facial reenactment, while revealing limitations in cross-dataset performance and probability calibration. Controlled comparisons will examine the contributions of separate representation learning, suppression of processing-related nuisance variation, and evidence-sensitive fusion across diverse datasets and processing conditions. Evaluation will assess detection, calibration, errors at fixed decision thresholds, and selective prediction to determine whether uncertainty helps distinguish reliable predictions from those requiring further review. Together, the studies seek to establish which facial representations provide transferable manipulation evidence and under what conditions their uncertainty can support more trustworthy use of deepfake detectors in human-supervised screening workflows.

CHAPTER 1

Introduction

A facial video communicates more than the appearance of a person. It records identity-related characteristics, facial expression, geometry, and changes across time. Generative models can alter these properties with different objectives: a face swap transfers identity, facial reenactment modifies a performance, and lip manipulation changes visible speech. The resulting videos can be visually convincing even when some relationships among these properties differ from those in genuine recordings. Deepfake detection therefore involves a representation problem as well as a classification problem. A detector must observe information that remains useful when the manipulation technique, recording conditions, or population differs from its training data.

FaceForensics++, Celeb-DF, and the DeepFake Detection Challenge support evaluation across manipulation and recording conditions [73, 56, 20]. Broader manipulation-family benchmarks such as DF40 further motivate examining where a detector succeeds, rather than relying on one aggregate score [92].

This dissertation investigates complementary facial representations and evidential learning as two connected means of addressing this challenge. Complementary representations seek useful information from different aspects of the face. Evidential learning provides a common representation for class support and the amount of evidence committed to a decision. The dissertation develops this investigation through two completed studies and a proposed third study with preliminary results. The sequence progresses from person-related signatures, to multi-view structural learning and uncertainty, and then to complementary image and video representations with evidence-sensitive integration.

1.1 Motivation

1.1.1 Generalization Beyond Familiar Manipulation Artifacts

A supervised detector can achieve high accuracy by exploiting any feature correlated with its training labels. Some such features reflect manipulation, while others reflect the way a dataset was assembled. Compression, crop boundaries, face size, background content, or a recurring synthesis pipeline may offer an easier learning signal than properties shared across manipulation methods. When the deployment distribution changes, the correlation can weaken or reverse. This is a form of shortcut learning [32], and its practical importance in deepfake detection motivates cross-dataset evaluation rather than reliance on within-dataset performance alone.

One response is to diversify training examples. Blending-based augmentation, broader generator coverage, and representation learning all provide ways to reduce dependence on a narrow artifact distribution [54, 77, 93]. Another response is to reconsider what the detector observes. Identity embeddings, facial behavior, and geometric relationships summarize properties that have meaning beyond a specific image generator. They may therefore provide a useful basis for transfer even though their extraction and predictive value remain sensitive to recording conditions. DBaGNet, the first study, examines this representation-centered approach through a joint descriptor and distance-based learning [26].

1.1.2 Uncertainty as an Empirical Property

A detector's class score and its reliability are related but distinct quantities. A high score for manipulation is not, by itself, evidence that similarly scored videos have the same empirical manipulation frequency in a new domain. Neural network calibration can differ from classification accuracy, and post-hoc calibration on one distribution does not automatically transfer to another [35]. For forensic use, it is also important to know whether uncertain predictions concentrate the detector's errors. This leads to two complementary questions: whether probabilities are calibrated, and whether uncertainty can identify decisions that merit further review.

Evidential deep learning assigns nonnegative evidence to classes and constructs a Dirichlet distribution over categorical probabilities [75]. The resulting representation contains a predicted probability and a measure of uncommitted evidence, commonly called vacuity. A multi-view system can additionally expose disagreement among representations. These quantities provide useful objects for analysis, but their interpretation depends on training and evaluation. Vacuity is not automatically an estimate of every source of uncertainty, and

disagreement does not identify which branch is correct. The dissertation evaluates their relationship to observed errors rather than treating their names as sufficient justification.

DISCERN, the second study, investigates this relationship using visual, semantic, and structural streams [27]. Its Inter-Branch Disagreement Calibration objective relates disagreement to the fused evidential output. The proposed third study retains the broader evidential perspective while examining a different representation set and explicit conflict redistribution. This progression creates a common question across studies: which information supports detection, and what can the accompanying uncertainty tell us about the resulting decisions?

1.1.3 Relevance to Forensic Reliability

Forensic reliability motivates evaluating false alarms, missed manipulations, input conditions, and reproducibility alongside average accuracy. NIST's scientific-foundation review places validation and the limits of digital investigation methods in this broader context [61]. The discussion associated with *Daubert* similarly motivates testable claims and documented error behavior [81]. This dissertation addresses those scientific properties through experiments; it does not evaluate courtroom admissibility.

1.2 Problem Formulation

Let $V = (x_1, \ldots, x_N)$ denote the ordered facial observations extracted from a video, with authenticity label $y \in \{0, 1\}$ for genuine and manipulated content, respectively. A detector produces a score $p(V)$ and a decision

$$\widehat{y}(V) = \mathbb{I}\{p(V) \geq \tau\}, \tag{1.1}$$

where τ is a threshold chosen using a designated development partition. The same threshold is then applied to evaluation data. A representation function may operate on individual frames, a sequence of frames, or a set of extracted facial signatures. A multi-view detector combines several such representations before or after classification.

Training examples are drawn from a source distribution $P_{\text{train}}(V, y)$, whereas the target distribution $P_{\text{test}}(V, y)$ may differ in manipulation methods, identities, recording conditions, or media processing. The practical objective is to retain useful discrimination and controlled error behavior under these changes. There is no assumption that every possible target distribution is covered by the available benchmarks. Generalization is studied through specified changes in data source, manipulation family, and image degradation.

For evidential models, the detector additionally returns quantities such as vacuity $u(V)$ and inter-view conflict $\mathcal{K}(V)$. Their usefulness can be assessed through calibration, error detection, and selective prediction. If $s(V)$ is an uncertainty score, a selective detector retains examples below a threshold γ:

$$a_\gamma(V) = \mathbb{I}\{s(V) \leq \gamma\}. \tag{1.2}$$

1.3 Research Questions

This dissertation investigates how complementary facial representations and evidential learning support generalizable deepfake video detection and reliable uncertainty estimation under distribution shift. Four research questions guide three connected studies, progressing from facial signatures and multi-view structural learning to spatio-temporal representation learning and reliability-aware evidence fusion.

RQ1. To what extent do identity, behavioral, and geometric signatures provide complementary evidence for detection across manipulation methods and datasets? *Addressed by DBaGNet* through feature-family ablations and cross-manipulation and cross-dataset evaluation.

RQ2. How does combining visual, semantic, and structural evidence through disagreement-aware evidential learning affect detection, calibration, and selective prediction under distribution shift? *Addressed by DISCERN* through stream and training-objective ablations, cross-dataset evaluation, and corruption, calibration, and selective-prediction experiments.

RQ3. Does integrating appearance, latent facial-structure, and video-dynamics representations in a three-branch spatio-temporal framework improve generalization beyond individual branches and two-branch combinations? *Addressed by CROSS-EXAM* through matched branch and branch-combination comparisons across datasets and manipulation families, temporal-order controls, and processing-nuisance suppression ablations.

RQ4. Can reliability-aware multisource conflict redistribution improve detection and uncertainty reliability compared with Dempster's rule when branches provide conflicting or weak evidence? *Addressed by CROSS-EXAM* through matched comparisons of reliability-discounted PCR6$^+$ and Dempster fusion, together with ablations of reliability discounting and one-sided disagreement regularization. Evaluation examines detection, calibration, error ranking, and selective prediction under distribution shift.

1.4 Research Scope and Contributions

This dissertation focuses on visual detection of manipulated facial videos, with particular emphasis on generalization and uncertainty reliability under distribution shift. Audio-only detection, attribution to a specific generator, formal proof of an individual prediction, and general-purpose synthetic-image detection are outside its scope. Spatio-temporal analysis examines the contribution of information across video frames through controlled experiments; it does not assume that learned video features identify physiological mechanisms or causal relationships governing facial motion.

The dissertation develops four contributions through three connected studies. The first, addressed by DBaGNet and aligned with RQ1, is a facial representation that combines identity, behavioral, and geometric signatures through metric learning. Feature-family ablations and cross-manipulation and cross-dataset experiments establish the contribution of these signatures within the evaluated conditions.

The second contribution, addressed by DISCERN and aligned with RQ2, is a multi-view framework integrating visual, semantic, and structural evidence through disagreement-aware evidential learning. Its completed evaluation examines detection, probability calibration, and selective prediction across datasets and processing conditions.

The proposed third contribution, aligned with RQ3, is CROSS-EXAM: Cross-Examining Independent Facial Evidence for Reliable Deepfake Video Detection. Its three-branch spatio-temporal framework combines PE-Core appearance features, CLIP-based latent facial structure, and V-JEPA 2 video-dynamics representations. Separate-view evidential training adapts each branch to its detection task. Matched single-branch, two-branch, and three-branch comparisons assess representation complementarity, while nuisance-suppression ablations and temporal-order controls isolate the contributions of processing robustness and temporal information.

The proposed fourth contribution, aligned with RQ4, is CROSS-EXAM's reliability-aware evidence-fusion framework. It combines development-selected reliability discounting with direct multisource PCR6$^+$ fusion and a second training stage that updates only the evidential heads using one-sided disagreement regularization. PCR6$^+$ is an established combination rule adopted from information-fusion research [19]. The contribution lies in integrating these components to combine complementary facial evidence while distinguishing committed disagreement from shared ignorance, and in establishing their effects on detection and uncertainty reliability. Comparisons with Dempster's rule and simpler fusion methods, together with separate discounting and training-objective ablations, assess the contribution of each component.

Table 1.1: The three studies and their roles in the dissertation. Completed-study findings retain their original evaluation protocols; the third study is presented as proposed work with preliminary evidence.

Study	Main representation	Main question	Evidence in the proposal
DBaGNet	Identity, behavior, geometry	Transfer of facial signatures	Published detection and ablation results
DISCERN	Visual, semantic, structural streams	Detection and uncertainty under shift	Accepted-paper results and methodological analysis
CROSS-EXAM	Separate appearance, latent facial structure, video	Complementarity and evidence-sensitive uncertainty	Preliminary reference; matched representation, suppression, and loss comparisons

Evaluation links these contributions to detection across datasets and manipulation families, incremental branch value, probability calibration, fixed-threshold errors, error ranking, and selective prediction. The experiments also examine trade-offs between generalization and uncertainty quality, establishing when additional representations improve detection and when uncertainty meaningfully identifies unreliable predictions.

1.5 Research Progress

DBaGNet has been published in IEEE Transactions on Computational Social Systems, and DISCERN has been accepted at the IEEE International Joint Conference on Biometrics in 2026. Chapters 3 and 4 present these completed studies. Chapter 5 specifies CROSS-EXAM and its prospective evaluation. Its preliminary results are identified with the earlier shared-CLIP reference configuration, which establishes feasibility and motivates the distinct appearance path.

The reference shows that video evidence contributes substantially on the evaluated reenactment families, while appearance has limited additional value when it shares an encoder with latent facial structure. CROSS-EXAM tests that representation bottleneck and the uncertainty limitation exposed by DISCERN. The video branch is interpreted as a source of video-level evidence; temporal-order effects are evaluated explicitly rather than assumed from the encoder architecture.

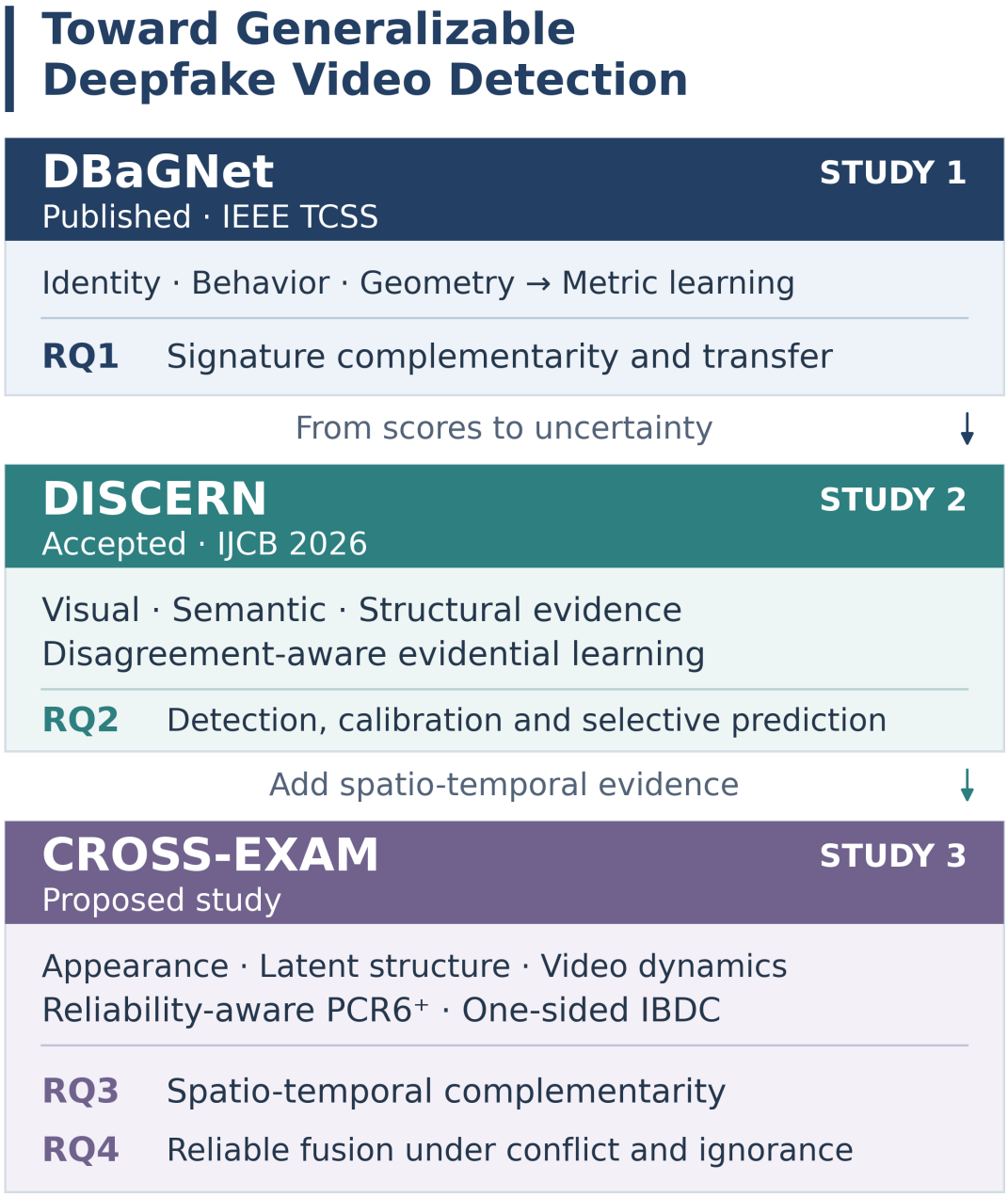


Figure 1.1: Dissertation roadmap linking three studies to four research questions: complementary facial signatures in DBaGNet (RQ1), multi-view evidential learning in DISCERN (RQ2), and spatio-temporal complementarity and reliability-aware evidence fusion in CROSS-EXAM (RQ3–RQ4). The progression connects generalizable detection with uncertainty estimation and fusion under conflicting or weak evidence.

1.6 Organization of the Proposal

Chapter 2 reviews facial manipulation detection, pretrained representations, evidential learning, and uncertainty assessment. Chapter 3 presents the facial-signature study, and Chapter 4 develops multi-view structural learning and disagreement-aware uncertainty. Chapter 5 presents CROSS-EXAM, its two-stage training design, the preliminary reference results, and the planned evaluation. Chapter 6 specifies the remaining tasks, tentative schedule, completion criteria, and expected contributions.

CHAPTER 2

Background and Related Work

Deepfake video detection draws on image forensics, facial analysis, representation learning, temporal modeling, and uncertainty estimation. These areas offer different answers to two questions: what information distinguishes authentic from manipulated content, and how should a detector express the reliability of its decision? This chapter reviews the approaches most relevant to the dissertation and develops the distinctions used in the subsequent studies. The emphasis is on the information a method observes, the assumptions that support its interpretation, and the evaluation needed to establish transfer beyond familiar data.

2.1 Facial Manipulation and the Detection Task

2.1.1 Manipulation Families

Face swapping, facial reenactment, and lip manipulation alter different properties of facial video. Face swapping primarily changes the depicted identity while attempting to preserve pose, expression, and scene context. Facial reenactment transfers or controls aspects of performance, including expressions and head movement. Lip manipulation changes visible articulation, often with the aim of consistency with a target utterance. Surveys of facial manipulation describe a broader range of generation pipelines, from graphics-based methods to neural synthesis [67, 83, 63].

The distinction between a manipulation family and a dataset is particularly important. A dataset may contain several families, and a family may appear in multiple datasets with different recording conditions. Family-resolved evaluation helps separate these effects. It can reveal that a detector's overall performance is dominated by its success on a common family while a smaller subset remains difficult. This is one reason the third study reports both source-domain and manipulation-family summaries.

2.1.2 The Processing History of a Facial Video

A digital video passes through acquisition, encoding, possible editing, resizing, and redistribution. Each stage can alter local texture, frequency content, and temporal sampling. Manipulation adds another transformation to that history, but benign processing can also create unusual patterns. A detector must consequently distinguish manipulation-related evidence from changes caused by compression, camera processing, or the analysis pipeline itself.

Fine residual statistics can provide useful forensic information, as illustrated by rich-model approaches and learnable residual filters [30, 37]. Their sensitivity is also a reason to document preprocessing. Resizing a face crop can remove a weak manipulation trace, while face alignment can introduce interpolation patterns. The presence of a trace in the original video does not guarantee its presence in the tensor supplied to a neural network. Conversely, a preprocessing artifact correlated with labels can become a shortcut.

2.2 Spatial and Artifact-Based Detection

2.2.1 Visual Classification and Local Inconsistencies

Early neural detectors established the value of convolutional visual representations for facial forgery detection. Compact architectures such as MesoNet and stronger image classification backbones could distinguish manipulated faces under benchmark conditions [1, 73]. Subsequent approaches sought features with a closer relationship to the manipulation process, including blending boundaries, local inconsistencies, and differences between facial regions.

Face X-ray learns a representation associated with blending, while self-blended images provide synthetic training examples intended to reduce reliance on a fixed collection of generators [54, 77]. These approaches illustrate a productive principle: a useful training task can expose structural features shared by several generation methods. Their transfer still depends on whether the target manipulation produces the relevant inconsistency and whether preprocessing preserves it. A generator that synthesizes an internally coherent face may not create the same boundary evidence as a compositing pipeline.

Local attention and region-sensitive methods further recognize that manipulated evidence is spatially uneven [100, 68]. Region-sensitive features can improve classification, although an attention map is not by itself a verified manipulation mask or causal explanation.

2.2.2 Frequency and Residual Representations

Frequency-based detection examines patterns that may be difficult to recognize in the image domain. Differences in upsampling, reconstruction, or compression can change spectral distributions. F3-Net studies frequency-aware clues, and related analyses investigate the frequency characteristics of generated images [71, 29, 24]. Residual representations suppress dominant image content to emphasize subtle differences, while spectral approaches can provide complementary information to semantic image features.

The usefulness of such representations depends on both generation and subsequent processing. A frequency pattern observed in training may be altered by re-encoding or resampling. It can also correlate with dataset origin rather than manipulation. Consequently, frequency evidence benefits from evaluation on genuine videos from multiple sources and under label-independent degradations. Applying a transformation only to manipulated examples would confound the target property with the intervention itself.

For this dissertation, frequency analysis is relevant in two ways. DISCERN includes forensic features alongside semantic observations in its structural stream. The proposed third study uses a manifold-based spatial representation and evaluates whether a separately parameterized appearance encoder adds useful information. Neither representation is interpreted as a universal fingerprint. Its role is assessed through detection, genuine-video behavior, and its incremental value in a system that already contains other strong views.

2.2.3 Reconstruction and Latent Facial Structure

Reconstruction-based detection learns a model of representations or image content and then uses reconstruction behavior for discrimination. The representation can include a residual magnitude, a residual direction, a latent code, or features learned jointly with classification. End-to-end reconstruction–classification learning demonstrates that reconstruction objectives can be incorporated into face forgery detection [10]. Variational autoencoders provide a probabilistic latent representation with a reconstruction objective and regularization toward a chosen prior [50, 42].

A reconstruction discrepancy is not intrinsically specific to manipulation. A genuine image from a different camera, pose range, or demographic distribution may also be difficult for a model trained on a restricted source. The training distribution and the reconstruction target therefore determine what a large residual means. A model trained only on genuine examples offers one interpretation: departure from the learned genuine distribution. A jointly trained discriminative reconstruction model offers another: a representation shaped by both reconstruction and class supervision. These alternatives should not be conflated in the

method description.

The variational autoencoder used for latent facial structure, labeled MR-VAE in the proposed architecture, characterizes latent facial structure within the current spatial system. Its value in the preliminary results is demonstrated by view-removal experiments. This evidence establishes a useful representation but does not imply that every correctly classified fake lies outside a mathematically identified manifold. CROSS-EXAM gives appearance a separate PE-Core feature path while retaining CLIP-based latent facial structure. This makes it possible to evaluate whether its redundancy arose partly from a shared spatial feature path.

2.3 Identity, Behavior, and Geometry

2.3.1 Identity-Related Representations

Face recognition models learn embeddings that preserve identity-related information across changes in pose, illumination, and expression. Margin-based objectives and quality-aware representations have advanced this task [18, 49]. For deepfake detection, such representations can expose relationships that are not captured by local texture alone. A face swap attempts to insert one identity into the motion, geometry, and context of another recording, creating an opportunity to study inconsistencies among these attributes.

Using a face-recognition embedding does not make detection an identity-verification task. A detector may lack an authentic reference image of the depicted person, and it may use identity features only as part of a learned representation. The key question becomes whether the embedding contributes transferable information about manipulation. DBaGNet investigates this through an identity signature combined with behavior and geometry. Its feature-family ablations are more informative about that contribution than a semantic interpretation of individual embedding dimensions.

2.3.2 Facial Behavior and Expression

Behavioral approaches study patterns in the way a face changes, including expressions, action units, and head movement. Work on soft biometric behavior illustrates that person-related motion can be useful for manipulated-video analysis [3, 2]. Visual speech representations provide another route to facial behavior, since articulation imposes structured changes around the mouth and adjacent regions [36].

Behavioral information may transfer when appearance artifacts change, but it introduces its own conditions. A short clip may contain little motion; an occluded mouth can compromise

speech features; a new speaking style or large head turn may differ from the training distribution. These conditions motivate evaluating the availability and quality of the observation, not just the class score. They also caution against interpreting every sequence-model gain as evidence for a specific physiological mechanism.

DBaGNet summarizes facial behavior using blendshape-related descriptors. DISCERN uses action units, expression, and pose in a semantic representation. The third study introduces a video encoder that observes an ordered sequence directly. These are distinct levels of temporal and behavioral modeling. A sequence of compact facial signatures emphasizes explicit measurements, whereas a pretrained video encoder can learn a broader latent representation. The dissertation studies their empirical contributions within their respective systems rather than assuming they measure the same property.

2.3.3 Geometric Relationships

Facial landmarks provide a compact description of shape and relative position. Geometric features can represent proportions, distances, and changes between facial regions. These measurements are attractive because their meaning is clearer than that of an arbitrary image feature, and they can complement identity and texture representations. However, projection, pose, expression, and landmark-estimation error all affect the measurements. A geometric relationship should therefore be interpreted in the coordinate system and normalization used by the extractor.

DBaGNet forms geometric profiles between upper and lower facial regions. The rationale is that manipulation can disturb relationships among identity, inner-face content, and outer-face structure. This is a testable representation hypothesis, not a universal rule of facial authenticity. Genuine faces exhibit substantial natural variation, and a generator may preserve many geometric relationships. The value of geometry is established by its contribution to detection after it is combined with the other signature families.

2.4 Pretrained Representations and Efficient Adaptation

2.4.1 Foundation Features for Forensic Transfer

Large pretrained encoders offer features learned from data beyond the relatively small collections used to train deepfake detectors. CLIP, for example, learns visual representations through image–text supervision [72]. Forensic adaptation then seeks to use that broad representation without over-specializing it to the artifacts of a single training collection.

Forensics Adapter, FSFM, Effort, and GenD illustrate different approaches to generalizable representation learning or adaptation [17, 86, 91, 98].

A pretrained model's generality does not eliminate the need for forensic evaluation. Its features can contain semantic or dataset information that is useful to a classifier but unrelated to manipulation. The adaptation objective determines which directions become discriminative. A useful experiment therefore compares the pretrained feature with the proposed residual, adapter, or auxiliary objective under a common head and training protocol. This comparison asks whether the proposed mechanism adds value beyond access to a strong backbone.

2.4.2 Low-Rank Adaptation and Staged Training

LoRA represents an update to a weight matrix through a low-rank factorization [43]. If $W_0 \in \mathbb{R}^{d_o \times d_i}$ is a pretrained matrix, an adapted layer can be written as

$$W = W_0 + sBA, \tag{2.1}$$

where $A \in \mathbb{R}^{r \times d_i}$, $B \in \mathbb{R}^{d_o \times r}$, r is the adapter rank, and s is a scaling factor. The pretrained matrix can remain frozen while the smaller factors are optimized. This reduces the number of trainable parameters, although the full pretrained layer is still executed during inference.

Staged training can make the effect of an added branch easier to measure. With spatial outputs fixed, the contribution of a new video representation can be evaluated without simultaneously changing the spatial feature space. Joint training offers a different possibility: gradients from the fused objective can coordinate the representations. Neither strategy is inherently superior. Joint optimization can improve task alignment, but it can also allow a dominant branch to carry the loss while another loses its independent predictive value.

Layer-norm adaptation provides another restricted update: it changes the scale and shift parameters of normalization layers while preserving the pretrained attention and feed-forward weights. DISCERN uses this strategy for its visual stream, and CROSS-EXAM uses it for PE-Core appearance. CROSS-EXAM then separates two optimization roles. Stage 1 learns each view through its permitted adaptation parameters and head; Stage 2 fixes those representations and coordinates only the three evidence heads. This makes a joint loss comparison possible without simultaneously changing the backbone features.

2.5 Temporal and Predictive Representations

2.5.1 Video Modeling for Deepfake Detection

Temporal detectors use relationships across observations rather than treating each frame as an independent image. Spatiotemporal networks, temporal coherence models, and visual-speech representations offer different ways to exploit these relationships [34, 102, 36]. Motion-magnified residual networks further investigate subtle temporal patterns in manipulated videos [64]. The literature supports temporal information as a useful source of evidence, while leaving open which representation is most complementary to a given spatial detector.

Temporal evaluation requires attention to sampling. A model given consecutive frames at a known frame rate observes local changes over physical time. A model given uniformly sampled crops across a long video receives an ordered but potentially sparse sequence. Both can be useful, but they support different claims. Sparse sampling may capture changes in pose, appearance, or expression without resolving fine motion. Temporal spacing and clip construction should therefore be reported alongside the model architecture.

An ordered-versus-shuffled comparison tests whether a trained model's predictions depend usefully on the supplied order. It does not alone prove a particular motion mechanism, because shuffling also changes the input distribution. A repeated-frame control removes between-frame variation while preserving a representative appearance. A retrained order-invariant head addresses yet another question: whether access to multiple observations, without order, accounts for the improvement. Together, these controls provide a more precise interpretation than the label "temporal branch."

2.5.2 Predictive and Contrastive Pretraining

Joint-embedding predictive architectures learn to predict representations rather than reconstruct pixels. I-JEPA predicts target-region embeddings from image context, whereas V-JEPA 2 learns a video representation through predictive self-supervision [4, 5]. For deepfake detection, the encoder features and the predictor's residuals are different candidate measurements. A useful encoder does not imply that its prediction error is specific to manipulation: unfamiliar genuine faces can also be difficult to predict.

Contrastive vision-language pretraining offers a different source of representations. Perception Encoder includes a core contrastive checkpoint, PE-Core, as well as separately aligned model variants [9]. CROSS-EXAM uses PE-Core for appearance at 448×448 resolution, with layer-norm adaptation. PE-Spatial is a separate aligned variant, rather than the appearance encoder used here.

The rationale for PE-Core is a change in the appearance feature path relative to the retained facial-structure view. Different pretrained weights and objectives create an opportunity for complementary information, but they do not imply statistically independent errors. A comparison must therefore measure both standalone detection and the additional value of appearance when the facial-structure and video opinions are already available. The two-view system is a necessary comparator even if the three-view system performs well in isolation.

A further issue is the information removed or preserved during adaptation. Compression and resizing change features even when the authenticity label is unchanged, and they can also destroy useful forgery evidence. Representation consistency should therefore be evaluated alongside detection rather than imposed as an unconditional requirement. CROSS-EXAM estimates processing-sensitive directions from paired clean and transformed training crops and suppresses their projection. Its evaluation compares this operation with ordinary augmentation, partial attenuation, and matched random directions under the same training exposure.

2.6 Evidential Learning and Multi-View Fusion

2.6.1 Dirichlet Evidence

For a C-class problem, an evidential network returns nonnegative evidence e_k and Dirichlet parameters $\alpha_k = e_k + 1$. The total strength is $S = \sum_{k=1}^{C} \alpha_k$, and the expected categorical probability is

$$p_k = \frac{\alpha_k}{S}. \tag{2.2}$$

The associated subjective opinion under a uniform base rate is

$$b_k = \frac{e_k}{S}, \qquad u = \frac{C}{S}, \qquad \sum_k b_k + u = 1. \tag{2.3}$$

This representation separates the predicted class balance from the amount of committed evidence [75]. In binary detection, equal evidence for the two classes gives a probability of one half regardless of whether the evidence is very small or very large. Vacuity distinguishes these cases even though the projected probability does not.

The distinction is mathematically useful, but it does not establish that the network's evidence magnitude is empirically calibrated. A training objective must shape that magnitude, and evaluation must determine whether it ranks errors or expresses meaningful confidence under shift. In particular, a high-evidence but incorrect prediction has low vacuity. Evidence-based uncertainty must therefore be assessed alongside class performance and other uncertainty

scores.

2.6.2 Agreement, Conflict, and Shared Errors

Multi-view evidential methods combine opinions from different observations or representations [39, 88, 8]. When views disagree, a fusion rule must determine how incompatible support affects the final opinion. When they agree, the rule must still account for the possibility that their errors are correlated. Statistical independence is not guaranteed by using separate heads, especially if the heads share a feature extractor or training data.

Shared ignorance is another important case. Two nearly uniform low-evidence opinions can agree because neither contains much information. An agreement-based regularizer that always reduces uncertainty can then encourage confidence without stronger support. Conversely, two highly committed opinions can disagree because one view observes a useful cue and the other follows a shortcut. Conflict identifies that incompatibility but does not specify which opinion is correct.

These cases motivate separating three concepts: uncertainty due to weak evidence, disagreement among committed views, and the predictive utility of a branch. DISCERN links disagreement to uncertainty through IBDC. The proposed study revisits that link using evidence-sensitive conflict and one-sided regularization. Its evaluation must include shared-ignorance and confidently wrong agreement cases, because aggregate calibration alone can conceal these distinct behaviors.

2.6.3 Discounting and Conflict Redistribution

Reliability discounting reduces the influence of a source judged less useful or less trustworthy. Its exact formulation matters. Scaling evidence before reconstructing an opinion changes belief and vacuity nonlinearly, whereas multiplying belief masses by a trust coefficient and moving the remainder to ignorance is a different operation. A method should state which quantity is transformed and at what point in the pipeline.

Dempster's normalized conjunctive rule combines compatible masses and normalizes by the surviving nonconflicting mass [76]. Proportional conflict redistribution assigns conflicting mass back to the propositions involved [78, 62, 19]. These are established alternatives with different behavior under conflict. CROSS-EXAM adopts direct multisource conflict redistribution as its integration operator and evaluates representation selection and evidential training as the methodological questions.

The current third study uses direct multisource PCR6^{+} because it provides a suitable interpretation of a view assigned zero evidence. A fully vacuous source does not change the

result, so suppressing a view corresponds to removing its information. Dempster's rule also has a vacuous identity when defined. The relevant distinction from direct PCR6 is how the universal set participates in conflict redistribution, not a claim that only PCR6$^+$ can ignore a vacuous source. Direct evaluation also avoids describing a nonassociative pairwise fold as a multisource operator.

2.6.4 Relationship to DiCoME and DISCERN

DiCoME applies multi-view evidential learning to deepfake detection using perception-oriented and artifact-oriented representations [48]. Its relevance is both architectural and experimental: it provides a strong spatial evidential reference and shows the value of a learned artifact representation. A spatial system can nevertheless have common weaknesses across its views when a manipulation family preserves the cues those views emphasize. This motivates testing a separately pretrained video representation and resolving results by manipulation family.

DISCERN studies a different decomposition into visual, semantic, and structural evidence. Its predicates and class-conditioned dependency models introduce additional descriptions of facial content, while IBDC connects disagreement to uncertainty. The preliminary reference uses spatial modules from a locally trained DiCoME-based system and an additional V-JEPA 2 video view. CROSS-EXAM extends this line of work through separate feature paths, processing-nuisance suppression, and joint evidential-head training. Its conceptual connection to DISCERN is the question of when disagreement should influence uncertainty; its feature decomposition and fusion operator differ.

2.7 Calibration, Error Detection, and Selective Prediction

2.7.1 Probability Calibration

A calibrated binary score relates predicted probabilities to observed event frequencies. Calibration can be assessed using reliability diagrams, expected calibration error, Brier score, and negative log-likelihood [35]. These metrics emphasize different properties. Binned calibration summaries depend on the binning convention, while proper scoring rules evaluate the full probability assignment. Reporting several complementary measures reduces dependence on one summary.

Calibration is distribution-dependent. A model calibrated on a balanced development set may encounter a different manipulation prevalence or a change in class-conditional scores. A post-hoc calibrator should therefore be trained on a designated partition and

evaluated unchanged on other data. Recalibrating separately on every test dataset answers an adaptation question, not the same cross-dataset question as fixed calibration.

2.7.2 Error Ranking and Abstention

An uncertainty score is useful for selective prediction if it ranks incorrect decisions above correct ones. Error-detection AUROC evaluates this ranking as a binary task, while risk–coverage curves show the tradeoff between retained workload and error rate. Excess area under the risk–coverage curve compares the observed ordering with an ideal ordering for the same set of predictions [31].

Good error ranking does not imply calibrated probabilities. A monotonic transformation can preserve ranking while changing numerical confidence substantially. Likewise, a detector with lower overall error can still have a poorer uncertainty ordering. For these reasons, selective prediction is evaluated together with full-coverage performance, and uncertainty scores are compared on the same predictions whenever the aim is to isolate the ranking mechanism.

2.8 Common Evaluation Definitions

The completed studies use their original protocols; their absolute scores should not be interpreted as a single common leaderboard. Across the dissertation, AUROC measures discrimination, whereas threshold errors and calibration describe different aspects of the predictions. Unless explicitly stated as frame-level, the evaluation unit is the video. AUC and AUROC denote the same receiver-operating-characteristic area. Chapters 3 and 4 report it as a percentage, preserving their published tables; Chapter 5 reports it on $[0, 1]$. Thus 95.72% equals 0.9572. Differences on the percentage scale are percentage points; probability-scale differences are absolute AUROC differences. Calibration errors are proportions unless a multiplier is printed. The number of classes is C, total conflict is $\mathcal{K}$, and pairwise conflict targets are κ.

For a decision threshold τ, false-positive rate and true-positive rate are

$$\mathrm{FPR}(\tau) = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}, \qquad \mathrm{TPR}(\tau) = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{2.4}$$

A threshold chosen on development data and transferred unchanged tests operating-point stability. Choosing a separate threshold on each test dataset instead describes adaptation to that dataset. Both can be informative, but they answer different questions.

For manipulated-class probabilities p_i and binary labels y_i, two probability-quality measures are

$$\text{NLL} = -\frac{1}{n}\sum_i [y_i \log p_i + (1 - y_i)\log(1 - p_i)], \tag{2.5}$$

$$\text{Brier} = \frac{1}{n}\sum_i (p_i - y_i)^2. \tag{2.6}$$

Expected calibration error compares confidence and correctness within bins. Its bin count and confidence definition are part of the metric specification. DISCERN uses ten equal-width bins; the preliminary third-study results use fifteen. Their ECE values are interpreted within those respective protocols.

For uncertainty score s_i, retain the set $\mathcal{S}_\gamma = \{i : s_i \leq \gamma\}$ and let z_i indicate an incorrect prediction. Selective coverage and risk are

$$\text{Coverage} = \frac{|\mathcal{S}_\gamma|}{n}, \qquad \text{Risk} = \frac{\sum_{i \in \mathcal{S}_\gamma} z_i}{|\mathcal{S}_\gamma|}. \tag{2.7}$$

Risk is considered only when at least one prediction is retained. Error-detection AUROC measures how well s_i ranks errors above correct predictions. These measures determine whether uncertainty is useful for deferral, while NLL and calibration assess the numerical probability assignments.

2.9 Evaluation Gaps and Dissertation Position

The reviewed literature motivates three connected gaps. First, high benchmark performance does not determine whether a representation transfers across manipulation families. Second, multiple views need not supply complementary errors, particularly when they share a feature path. Third, an evidential uncertainty output requires empirical validation beyond its mathematical definition. These gaps are addressed by the sequence of studies rather than by claiming a single architecture resolves every aspect of forensic reliability.

DBaGNet studies whether meaningful facial signatures improve transfer. DISCERN studies how heterogeneous evidence and disagreement-aware learning affect detection and uncertainty. The third study examines the additional information provided by a video representation, the contribution of latent facial structure, and the behavior of evidence discounting and conflict. CROSS-EXAM tests these questions through a separate appearance path with nuisance suppression and a second stage that trains only the evidential heads with one-sided IBDC.

Architecture specifies what a model can observe; experiments establish what it contributes. The study chapters connect these distinctions to their experiments. Chapter 5 specifies the matched comparisons and prospective evaluation for the remaining research.

CHAPTER 5

CROSS-EXAM: Complementary Facial Evidence and Reliable Uncertainty

The first two studies establish the value of facial representations and extend detection to uncertainty assessment. They also expose two questions that a multi-view detector must answer. Does each view contribute information that the other views miss? When their predictions agree, is that agreement supported by evidence or does it reflect shared ignorance? CROSS-EXAM addresses these questions through separate appearance, latent facial-structure, and video representations, followed by evidence-sensitive joint training.

RQ3. Can separately pretrained facial representations and processing-nuisance suppression improve complementary manipulation evidence, and can one-sided disagreement calibration improve uncertainty without treating weak agreement as a reason for certainty?

5.1 Rationale and Research Hypotheses

A detector can contain several heads while depending on one feature path. Distinct classification objectives do not necessarily overcome this shared information bottleneck. The preliminary reference in this chapter illustrates the issue: adding a video encoder substantially improves reenactment detection, but its appearance head adds little to the latent facial-structure–video pair. Appearance receives zero reliability in three of five coefficient-fitting repetitions. These findings motivate a separate appearance encoder and a direct test of its incremental value.

CROSS-EXAM combines three views. The appearance view (APP) adapts PE-Core at 448×448 resolution through layer-normalization parameters and suppresses feature directions associated with benign image processing. The latent facial-structure view (LFS) uses a separately adapted CLIP encoder and an MR-VAE. The video view (TD) uses V-JEPA 2 with low-rank adaptation. Here, *independent* denotes separate parameterized feature paths and separate first-stage supervision. It does not assume statistically independent errors

or independent pretraining data. TD retains the temporal/disagreement branch acronym; it denotes a video representation, and its contribution is not attributed to temporal order without an order-specific test.

The first hypothesis is that the separate APP path contributes useful information beyond LFS and TD, particularly when nuisance suppression reduces sensitivity to processing conditions. The second is that a one-sided disagreement objective can improve uncertainty by responding to committed conflict without rewarding certainty during shared ignorance. The evaluation separates backbone choice, nuisance suppression, and disagreement training so that an improvement can be assigned to the component that produced it.

5.2 CROSS-EXAM Architecture

5.2.1 Overview

Let $V = (x_1, \ldots, x_N)$ denote ordered facial crops from a video, with label $y \in \{\mathrm{G}, \mathrm{F}\}$. The three branches are indexed by $v \in \{A, L, T\}$ for APP, LFS, and TD. APP and LFS produce frame-level opinions that are averaged within a video. TD produces one opinion from an ordered clip. Their video-level evidence is discounted using fixed reliability coefficients and integrated by direct multisource PCR6$^+$ [19]. The outputs are an authenticity score, fused vacuity, and retained conflict. Figure 5.1 shows the inference path; Figure 5.2 separates the two training stages.

5.2.2 Appearance Evidence and Nuisance Suppression

PE-Core provides a feature path separate from the CLIP representation used by LFS [9]. APP uses 448×448 facial crops and a global image embedding $z_A = f_{\mathrm{PE}}(x)$. During separate-view training, only PE-Core's layer-normalization parameters are adapted; the other pretrained encoder parameters remain fixed. A 256-unit projection with GELU activation and a two-output classifier map the suppressed feature to nonnegative genuine and manipulated evidence. The projection is part of the APP evidential head.

Compression and resampling can change an embedding without changing the authenticity label. Nuisance suppression targets these processing directions using paired observations of the same training crop. Let a be a label-independent processing transformation and let

$$\widetilde{z}_A(x) = \mathrm{diag}(\sigma + \epsilon)^{-1}\big(z_A(x) - \mu\big), \qquad \delta(x, a) = \widetilde{z}_A(a(x)) - \widetilde{z}_A(x), \tag{5.1}$$

where μ and σ are feature means and standard deviations estimated from the training

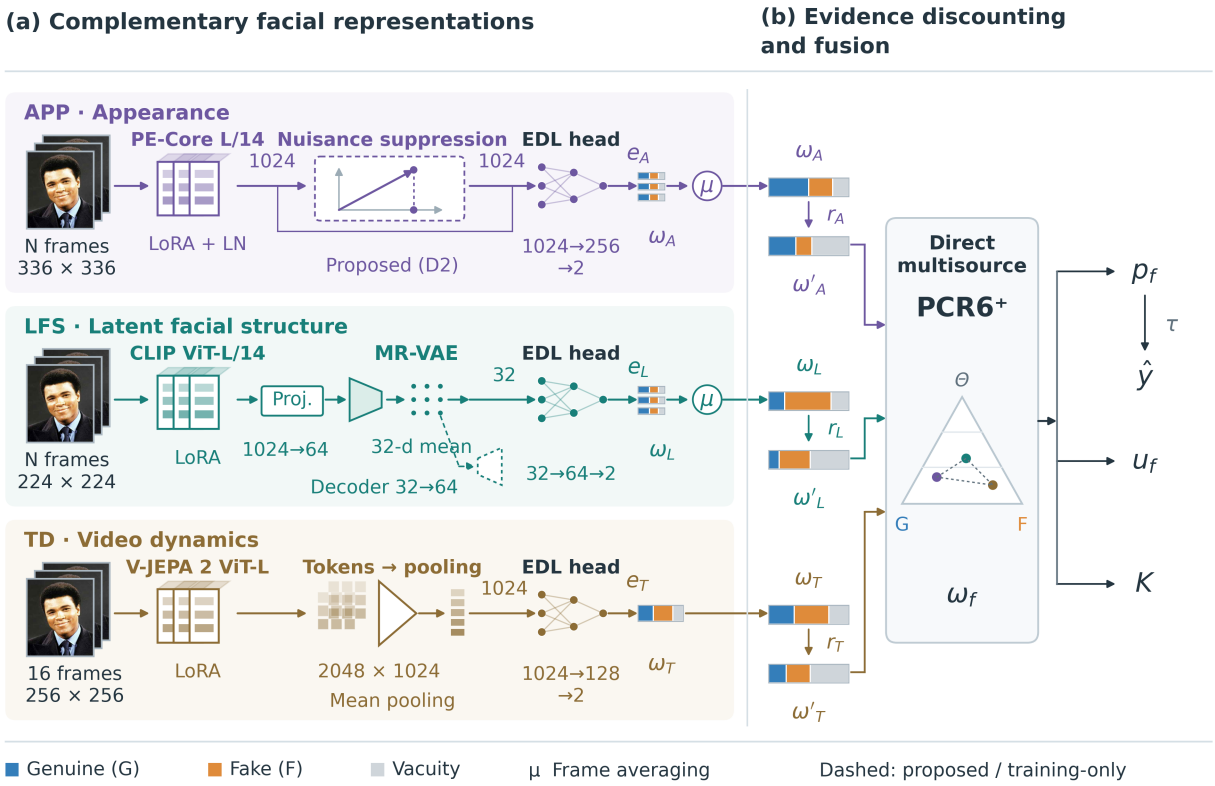


Figure 5.1: CROSS-EXAM architecture. APP uses PE-Core with layer-norm adaptation and nuisance suppression; LFS uses CLIP with LoRA and a 32-dimensional MR-VAE latent; TD uses V-JEPA 2 with LoRA and pooled clip tokens. Spatial opinions are averaged before video-level evidence discounting. Direct multisource PCR6$^+$ integrates the adjusted opinions, retaining both vacuity and conflict for analysis. Opinion bars and the simplex are schematic illustrations, not measured predictions. Training status is specified separately in Fig. 5.2.

partition, and $\epsilon > 0$ provides numerical stability. Stacking the paired differences as rows gives a matrix D. Its leading q right singular vectors form an orthonormal basis Q_q. APP removes the projection onto that basis:

$$z'_A = (I - Q_q Q_q^{\mathsf{T}})\widetilde{z}_A, \qquad e_A(x) = \mathrm{softplus}\Big(h_A(z'_A(x))\Big). \tag{5.2}$$

The basis estimates directions sensitive to the chosen transformations; it is not assumed to identify every nuisance factor. Processing can also erase real forensic information, so the retained rank is selected by detection and robustness on development data, rather than by reconstruction of processing variation alone.

The first-stage APP schedule uses an EDL warmup to adapt the layer norms and head. It then estimates μ, σ, and Q_q from paired training features and holds them fixed while continuing layer-norm and head adaptation with Eq. (5.2). The same fixed transform is used in Stage 2 and at inference. This makes the suppression operator reproducible and prevents evaluation samples from influencing its basis. The principal controls use the same paired exposures with no projection, partial attenuation $(I - \rho Q_q Q_q^{\mathsf{T}})$ for $0 < \rho < 1$, and a random orthonormal basis of matched rank. These comparisons distinguish the effect of the selected directions from augmentation and generic dimensionality reduction.

5.2.3 Latent Facial-Structure Evidence

LFS uses CLIP ViT-L/14 features from 224×224 crops, with LoRA applied to attention query and value projections [72, 43]. Its pretrained weights remain fixed. An MR-VAE maps the adapted feature to a 32-dimensional latent representation with a reconstruction objective and KL regularization [42]. The evidential head uses the latent mean at inference:

$$z_L(x) = \mu_{\mathrm{VAE}}(f_{\mathrm{CLIP}}(x)), \qquad e_L(x) = \mathrm{softplus}\Big(h_L(z_L(x))\Big). \tag{5.3}$$

A 64-unit hidden layer maps the latent representation to two evidence outputs. The decoder provides auxiliary supervision during Stage 1. Stage 2 updates the LFS evidential head while freezing the CLIP adapters and the entire MR-VAE, including its latent encoder. Thus joint uncertainty training changes how latent structure supports classification without changing the latent representation itself.

The reconstruction objective encourages a structured description of facial features alongside discrimination. It does not establish a universal genuine-face manifold or prove that every manipulated face lies outside it. Its empirical role is tested by removal and by replacing MR-VAE with a parameter-matched discriminative head. The branch builds on the manifold-

Table 5.1: CROSS-EXAM branch specification. Hidden projections belong to the evidential heads. All three heads produce two nonnegative evidence values; frame opinions are pooled only for APP and LFS.

View	Input	Feature path	Evidence head
APP	448×448 crop	PE-Core; layer-norm adaptation; nuisance projection	256-unit projection; GELU; two outputs
LFS	224×224 crop	CLIP ViT-L/14; LoRA; MR-VAE latent of size 32	64-unit hidden layer; two outputs
TD	16 crops, 256×256	V-JEPA 2 ViT-L; LoRA; token mean	Layer norm; 256-unit projection; GELU; two outputs

based spatial formulation used in DiCoME [48], while CROSS-EXAM separates its encoder from APP.

5.2.4 Video Evidence

TD processes a clip $c(V) = (x_{i_1}, \ldots, x_{i_{16}})$ sampled uniformly in temporal order over available facial crops. Each crop uses 256×256 video-encoder preprocessing. V-JEPA 2 ViT-L supplies clip tokens; attention query and value projections receive LoRA updates while pretrained weights remain fixed [5]. Mean-pooled tokens enter a head with layer normalization, a 256-unit projection, GELU activation, and two softplus outputs:

$$e_T(V) = \mathrm{softplus}\Big(h_T(\mathrm{pool}(f_{\mathrm{VJ}}(c(V))))\Big). \tag{5.4}$$

The trainable projection belongs to the TD head in both training stages. LoRA is updated only in Stage 1. Sampling is over stored facial observations, so the clip may span a different physical duration for different videos. Videos with fewer than 16 usable crops repeat uniformly selected indices in order; videos without a usable crop are recorded as processing failures and excluded from score-based metrics with coverage reported separately.

The video representation can capture relationships across observations that frame-independent heads cannot observe. This capability does not establish which relationships drive detection. Input shuffling, repeated-frame clips, and a trained order-invariant baseline test the contribution of ordering and multiple observations in Section 5.6.

5.2.5 From Evidence to Video-Level Opinions

The authenticity frame is $\Theta = \{\mathrm{G}, \mathrm{F}\}$. Following evidential classification [75], a view's nonnegative evidence $e_{v,k}$ defines Dirichlet parameters $\alpha_{v,k} = e_{v,k} + 1$ and total strength $S_v = \alpha_{v,\mathrm{G}} + \alpha_{v,\mathrm{F}}$. With a uniform base rate, the corresponding opinion is

$$b_{v,k} = \frac{e_{v,k}}{S_v}, \qquad u_v = \frac{2}{S_v}, \qquad p_{v,k} = b_{v,k} + \frac{u_v}{2}. \tag{5.5}$$

Here $b_{v,k}$ is the belief assigned to class k, u_v is vacuity, and $p_{v,k}$ is the projected class probability. The opinion satisfies $b_{v,\mathrm{G}} + b_{v,\mathrm{F}} + u_v = 1$. A vacuous opinion has $u_v = 1$ and no committed class evidence.

For each spatial view, opinions are averaged across the available crops:

$$\begin{aligned} \bar{b}_{v,k} &= \frac{1}{N}\sum_{n=1}^{N} b_{v,k}(x_n), \\ \bar{u}_v &= \frac{1}{N}\sum_{n=1}^{N} u_v(x_n), \qquad v \in \{A, L\}. \end{aligned} \tag{5.6}$$

Averaging preserves the opinion constraint. The corresponding video-level evidence is recovered as $\bar{e}_{v,k} = 2\bar{b}_{v,k}/\bar{u}_v$. The video branch already supplies one clip-level opinion. The following equations use $b_{v,k}$ and u_v for these video-level quantities.

5.2.6 Reliability-Weighted Evidence

Representations may differ in both discrimination and evidence magnitude. A coefficient $r_v \in [0, 1]$ for each view scales its evidence before fusion:

$$e'_{v,k} = r_v e_{v,k}. \tag{5.7}$$

Equivalently, letting $d_v = r_v(1 - u_v) + u_v$, the adjusted opinion is

$$b'_{v,k} = \frac{r_v b_{v,k}}{d_v}, \qquad u'_v = \frac{u_v}{d_v}. \tag{5.8}$$

Thus $r_v = 1$ retains the original opinion, whereas $r_v = 0$ yields a vacuous source. Intermediate coefficients reduce evidence strength while preserving the underlying class-support ratio. This operation scales evidence; it differs from linear discounting of belief masses.

The coefficients are selected on source-grouped development partitions, as specified in Section 5.6, and remain fixed during joint head training and inference. They regulate the

contribution of each representation at the model level. Variation across individual videos enters through the predicted evidence, rather than through a sample-dependent reliability network.

5.2.7 Direct Multisource Conflict Redistribution

The adjusted opinions are combined using PCR6$^+$ [19]. The binary frame permits a compact formulation. Each source assigns masses

$$m_v(\{\mathrm{G}\}) = b'_{v,\mathrm{G}}, \quad m_v(\{\mathrm{F}\}) = b'_{v,\mathrm{F}}, \quad m_v(\Theta) = u'_v. \tag{5.9}$$

Let $X = (X_A, X_L, X_T)$ range over tuples with $X_v \in \{\{\mathrm{G}\}, \{\mathrm{F}\}, \Theta\}$, and define the tuple mass $q(X) = \prod_v m_v(X_v)$. A tuple is conflicting when $\bigcap_v X_v = \emptyset$; let $\mathcal{C}$ contain these tuples. On the binary frame, a conflicting tuple necessarily contains both singleton classes.

For class k, the nonconflicting mass is

$$c_k = \prod_v (b'_{v,k} + u'_v) - \prod_v u'_v, \qquad u_\cap = \prod_v u'_v. \tag{5.10}$$

For a conflicting tuple, let $w_k(X) = \sum_{v:X_v=\{k\}} m_v(\{k\})$ denote singleton support. Direct PCR6$^+$ returns conflict to the involved singleton classes:

$$b_{f,k} = c_k + \sum_{X \in \mathcal{C}} q(X) \frac{w_k(X)}{w_{\mathrm{G}}(X) + w_{\mathrm{F}}(X)}, \tag{5.11}$$

$$u_f = \prod_v u'_v. \tag{5.12}$$

Zero-mass tuples are omitted. The resulting masses satisfy $b_{f,\mathrm{G}} + b_{f,\mathrm{F}} + u_f = 1$. A source with $r_v = 0$ is vacuous and does not change the fused opinion. The rule is evaluated jointly over the sources rather than by a sequential pairwise fold, since proportional conflict redistribution is not generally associative. This is an adopted integration rule [19]; the proposed contribution concerns representations and uncertainty training.

5.2.8 Prediction, Vacuity, and Conflict

The authenticity score is the projected fake probability

$$p_f = b_{f,\mathrm{F}} + \frac{u_f}{2}. \tag{5.13}$$

The conflict mass before redistribution is retained as

$$\mathcal{K} = \sum_{X \in \mathcal{C}} q(X) = 1 - c_{\mathrm{G}} - c_{\mathrm{F}} - u_{\cap}. \tag{5.14}$$

The output $(p_f, u_f, \mathcal{K})$ separates the predicted class from two aspects of the evidential state. Vacuity summarizes remaining uncommitted mass, whereas conflict records incompatible class support across views. Since Eq. (5.12) is a product, strongly committed views can yield low vacuity even when they disagree. Retaining $\mathcal{K}$ makes this disagreement available for analysis rather than interpreting vacuity as a complete description of uncertainty.

Vacuity is evaluated for selective prediction and its error ranking is compared with conflict and predictive entropy. The decision threshold is selected on FF++ validation scores and then kept fixed for evaluation. No benchmark-specific decision threshold is introduced.

5.3 Two-Stage Evidential Training

CROSS-EXAM separates learning each view from coordinating the evidential heads. Stage 1 gives every branch its own classification supervision. Stage 2 keeps the learned feature paths fixed and jointly adjusts the three evidential heads using supervised EDL and one-sided IBDC. Figure 5.2 identifies the updated modules in each stage.

5.3.1 Stage 1: Separate View Training with EDL

For branch evidence e_v and one-hot label y, let $\alpha_v = e_v + \mathbf{1}$. The per-view objective combines the Dirichlet type-II negative log-likelihood with annealed KL regularization [75]:

$$\begin{aligned}\mathcal{L}_{\mathrm{EDL}}(e_v, y) = &\sum_{k \in \Theta} y_k \Big(\log S_v - \log \alpha_{v,k} \Big) \\ &+ \beta(t)\ \mathrm{KL}[\mathrm{Dir}(\widetilde{\alpha}_v) \parallel \mathrm{Dir}(\mathbf{1})]\,,\end{aligned} \tag{5.15}$$

where $\widetilde{\alpha}_v = y + (\mathbf{1} - y) \odot \alpha_v$ suppresses the target-class evidence in the KL term and $\beta(t)$ is an annealing coefficient. APP and TD minimize their respective EDL losses. LFS additionally minimizes feature reconstruction error and the variational latent KL term:

$$\mathcal{L}_L^{(1)} = \mathcal{L}_{\mathrm{EDL}}(e_L, y) + \lambda_{\mathrm{rec}} \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{VAE}} \mathcal{L}_{\text{VAE-KL}}. \tag{5.16}$$

The VAE KL term regularizes the latent distribution; it is distinct from the Dirichlet KL term that regulates class evidence. APP and LFS receive frame supervision, while TD receives

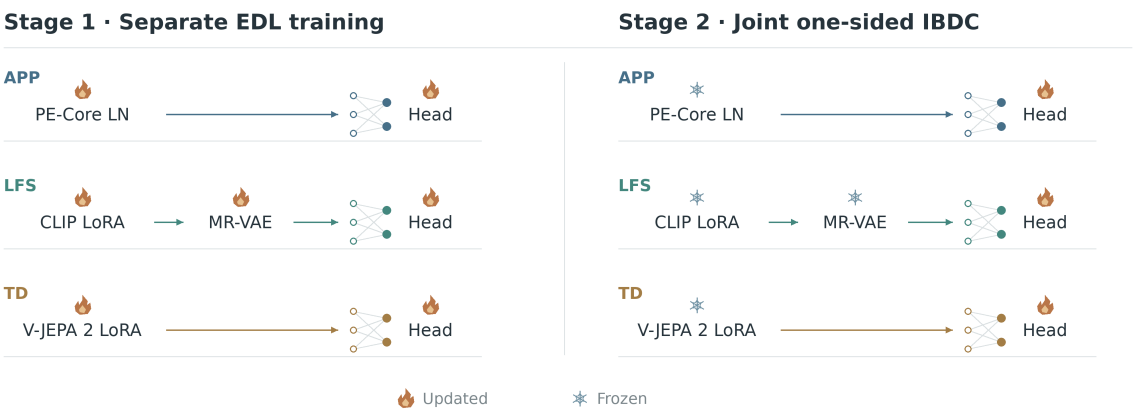


Figure 5.2: Stage-specific training in CROSS-EXAM. Stage 1 separately updates PE-Core layer norms and the APP head; CLIP LoRA, MR-VAE, and the LFS head; and V-JEPA 2 LoRA and the TD head. Stage 2 jointly updates the three evidential heads only. The pretrained weights, layer-norm adaptation, low-rank adapters, MR-VAE, and nuisance transform are fixed in Stage 2. Head projections remain trainable because they are part of the evidential heads.

video supervision. Every branch learns without a loss that matches another branch's class probability.

The updated parameters are PE-Core layer norms and the APP head; CLIP LoRA, MR-VAE, and the LFS head; and V-JEPA 2 LoRA and the TD head. All other backbone weights remain fixed. Processing augmentations are label-independent and consistent within a TD clip. The same source-grouped training partition is used for the three views so that representation differences are not confounded with different labeled video sets.

5.3.2 Stage 2: Joint Head Training with One-Sided IBDC

After separate-view training, the feature extractors, adaptation parameters, MR-VAE, and nuisance transform are frozen. The three evidential heads are optimized together on video-level opinions. Reliability coefficients are fitted after Stage 1 and then held fixed through Stage 2 and inference. A matched no-IBDC run uses the same frozen representations, coefficients, head initialization, and training schedule.

Original IBDC aligns fused vacuity with disagreement between normalized class probabilities. If every branch produces nearly uniform probabilities, disagreement can be small even though all opinions are weak. In DISCERN's symmetric binary-cross-entropy formulation, $d = 0$ reduces the disagreement loss to $-\log(1-u)$, which favors decreasing u. Shared ignorance can therefore supply an incentive for certainty. One-sided IBDC changes both the

target and the direction of the penalty.

Using the discounted opinions that enter fusion, committed pairwise conflict is

$$\kappa_{ij} = b'_{i,\mathrm{G}} b'_{j,\mathrm{F}} + b'_{i,\mathrm{F}} b'_{j,\mathrm{G}}, \qquad 0 \le \kappa_{ij} \le (1 - u'_i)(1 - u'_j). \tag{5.17}$$

The upper bound follows because the two cross-class products are a subset of all products in $(b'_{i,\mathrm{G}} + b'_{i,\mathrm{F}})(b'_{j,\mathrm{G}} + b'_{j,\mathrm{F}})$. Conflict vanishes when either opinion is vacuous. The detached target is

$$\kappa(V) = \mathrm{sg}\Big(\max_{i<j} \kappa_{ij}(V)\Big), \tag{5.18}$$

where sg blocks gradients through the target. One-sided IBDC is

$$\mathcal{L}_{\mathrm{IBDC+}} = \mathbb{E}_V \left[\max\{0, \kappa(V) - u_f(V)\}\right]. \tag{5.19}$$

It penalizes fused vacuity below the committed-conflict target. When $u_f \ge \kappa$, the term is inactive: it does not reward reducing vacuity merely because the branches agree. In particular, fully vacuous opinions have $\kappa = 0$ and receive no certainty incentive from this term. The pairwise maximum κ is a training target; it differs from the total multisource conflict $\mathcal{K}$ retained at inference in Eq. (5.14).

Figure **??** isolates the directional behavior of the regularizers.

Stage 2 retains per-view supervision so that increasing uncertainty cannot become the sole training objective:

$$\mathcal{L}^{(2)} = \frac{1}{3} \sum_{v \in \{A,L,T\}} \mathcal{L}_{\mathrm{EDL}}(\bar{e}_v, y) + \lambda_+ \mathcal{L}_{\mathrm{IBDC+}}, \tag{5.20}$$

where $\bar{e}_A$ and $\bar{e}_L$ are recovered from pooled frame opinions, and $\bar{e}_T = e_T$. The optimization variables are exactly the APP, LFS, and TD head parameters, including their internal projections. There is no gradient update to PE-Core layer norms, either LoRA module, or MR-VAE in this stage.

The loss removes a specific certainty incentive; it does not guarantee calibrated probabilities. Since $u_f = \prod_v u'_v$, increasing fused vacuity can reduce evidence in several views. Detachment prevents direct optimization of the conflict target but does not prevent indirect changes in class probabilities as heads change. A correct specialist can also disagree with two incorrect views. The evaluation therefore checks branch contribution, high-conflict correct predictions, shared ignorance, and confidently wrong agreement, rather than interpreting every conflict as an error.

5.4 Preliminary Reference and Experimental Protocol

5.4.1 Reference Configuration

The preliminary reference extends a locally trained DiCoME-based spatial system [48] with a V-JEPA 2 video view. Its appearance and latent facial-structure heads share a CLIP ViT-L/14 feature path. The appearance projection has 64 units, the MR-VAE latent has 32 units, and the video head has 128 units. The spatial checkpoint is held fixed while the video adapters and head are trained. The reference's off-manifold label corresponds to LFS and is denoted L in the tables here. The reference uses the opinion pooling, evidence scaling, and direct multisource fusion defined above, but does not include the PE-Core appearance path, nuisance projection, or Stage 2 one-sided IBDC.

All numerical results in Section 5.5 belong to this reference. They establish the empirical motivation for CROSS-EXAM: strong spatial–video complementarity, a useful latent facial-structure view, and limited additional value from appearance when it shares the spatial encoder. The proposed architecture is evaluated through the matched experiments in Section 5.6.

5.4.2 Reference Training and Model Selection

The reference detector was trained on FaceForensics++ (FF++) c23 training videos [73]. The video branch uses class-balanced sampling and an evidential classification objective with annealed KL regularization. LoRA adapters of rank 8, scaling factor 16, and dropout 0.1 are applied to attention query and value projections. Training uses AdamW with weight decay 0.01, a batch size of 32, and ten training epochs. The adapter and head learning rates are 2×10^{-4} and 2×10^{-3}, respectively, with a cosine schedule ending at 10^{-6}. Spatial inputs are resized to 224×224, and video inputs use the video encoder's 256×256 preprocessing.

Training uses label-independent image augmentations applied consistently within a clip. An unmodified clip is selected with probability 0.40; JPEG compression at quality 70, 50, or 30 and down–up resampling at scale 0.75, 0.50, or 0.35 each have probability 0.10. The same crop exports are used across evaluated methods, with video aggregation appropriate to each method.

Model selection uses FF++ validation and the two validation mixtures in the experimental protocol. The selection objective averages AUROC over swap, reenactment, and in-the-wild groups. Reliability coefficients are selected from $\{0, 0.25, 0.5, 0.75, 1\}$, subject to $\max_v r_v = 1$, using five repeated source-grouped partitions. Related manipulations and augmented variants of a source video remain within the same partition. These repetitions vary coefficient

estimation conditional on a trained detector; reported fusion results average the metrics over the resulting five coefficient sets. View subsets and fusion alternatives each undergo the same coefficient-selection procedure using their own outputs.

Published DISCERN and local re-evaluation. The DISCERN comparator in this chapter is a fixed locally retained detector checkpoint re-evaluated on the common crop exports and video-level scoring pipeline. The laboratory identifier R0 refers to that checkpoint; "replay" means inference from saved weights, not retraining or a three-seed average. Chapter 4 instead reports the accepted study's three-seed results with 32-frame video averaging. The two evaluation records are therefore kept distinct. For CDF-v2, CDF-v3, and DFDCP, the published results are 0.9572, 0.9427, and 0.9435, whereas the local checkpoint gives 0.9509, 0.8954, and 0.9064. The local contrast at CDF-v3 is $0.9639 - 0.8954 = 0.0685$; the numerical gap to the published mean is only $0.9639 - 0.9427 = 0.0212$ and is not a matched-protocol improvement. This distinction prevents checkpoint and evaluation differences from being attributed to the proposed representation.

5.4.3 Benchmarks and Comparators

The cross-dataset suite contains CDF-v2 [56], CDF-v3 [57], DFD [23], DFDC [20], DFDCP [21], DeepfakeEval-2024 [11], FFIW [104], KoDF [52], IDForge [89], and DFDM [46]. Table 5.2 reports all ten benchmarks in two panels. Its mean weights the datasets equally.

Manipulation-family analysis uses the available face-swap and reenactment subsets of DF40 [92]. This evaluation comprises 41 generator–source entries: 19 face-swap entries and 22 reenactment entries across Celeb-DF and FF++ source domains. A generator is evaluated against genuine videos from its corresponding source domain. The primary summary weights the four family–source cells equally; a generator-macro summary averages all 41 entries. These are different aggregation conventions and are identified explicitly.

Comparators include DISCERN, DiCoME, GenD variants, FSFM, Effort, and Forensics Adapter [27, 48, 98, 86, 91, 17]. Values are obtained by local evaluation on common video exports rather than copied from published tables. The DISCERN row denotes the reference checkpoint used in this evaluation. Additional comparisons vary the view subset and the fusion rule while preserving the underlying representation outputs.

5.4.4 Metrics and Additional Analyses

Detection performance is measured by video-level AUROC. For threshold-dependent analyses, each evaluated system uses its equal-error-rate threshold on FF++ validation, fixed across

test benchmarks. The evaluation reports expected calibration error (ECE) using 15 equal-width bins and negative log-likelihood (NLL). Error-detection AUROC measures how well an uncertainty score ranks incorrect decisions above correct ones. Selective prediction retains videos with lower uncertainty; risk is their classification error rate. Excess area under the risk–coverage curve (E-AURC) measures the difference from an optimal error ordering [31].

Additional evaluations examine JPEG compression at quality 40 and down–up resampling at scale 0.60. These assess different severities of the augmentation types used during training. The evaluation also covers four detector configurations: two video-branch seeds paired with the primary spatial checkpoint, and two matched spatial and video seeds using an alternative spatial training recipe. The configuration variations are summarized with the preliminary findings below.

5.5 Preliminary Results and Analysis

5.5.1 Cross-Dataset Detection

Table 5.2 shows that the measured reference is particularly effective on CDF-v3, KoDF, and DeepfakeEval-2024, attaining AUROCs of 0.9639, 0.9600, and 0.7235, respectively. These are the highest values among the locally evaluated systems on those benchmarks. Over the complete ten-benchmark suite, the framework reaches a mean AUROC of 0.9129, compared with 0.8959 for the DISCERN reference and 0.8945 for GenD-CLIP, the two strongest comparator means in this evaluation.

The distribution of gains is more informative than the aggregate alone. On CDF-v3, combining only the spatial views gives 0.8810, whereas the full system reaches 0.9639. On KoDF, the corresponding values are 0.8547 and 0.9600. These within-framework comparisons connect improved detection to the inclusion of video evidence under the same fusion and coefficient-selection procedure. The reference underperforms the local DISCERN comparator on DFDC (0.8530 versus 0.8879) and ForAda on DFDCP (0.8597 versus 0.9289), deficits of 0.0349 and 0.0692 AUROC. GenD-DINO also leads on FFIW. These are material weaknesses, particularly for evaluation on diverse real-world videos. The results thus favor complementary representations across the suite rather than a uniformly dominant representation on every dataset.

5.5.2 Complementarity Across Manipulation Families

Table 5.3 separates DF40 performance by family and source domain. The spatial pair is strong on face swapping, with AUROCs of 0.9674 and 0.9709 on the Celeb-DF- and FF++-sourced

Table 5.2: Measured reference: ten-benchmark comparison of video-level AUROC, shown in two panels. The DISCERN row uses the fixed local checkpoint, not the published three-seed mean. All rows use the local evaluation protocol described in the text.

Method	CDF-v2	CDF-v3	DFD	DFDC	DFDCP
DISCERN (local checkpoint)	0.9509	0.8954	**0.9431**	**0.8879**	0.9064
GenD-CLIP [98]	0.9534	0.8458	0.9350	0.8708	0.9162
DiCoME [48]	**0.9693**	0.8691	0.9343	0.8818	0.8755
FSFM [86]	0.9554	0.8751	0.9219	0.8844	0.9061
GenD-DINO [98]	0.9177	0.8172	0.9165	0.8470	0.9114
GenD-PE [98]	0.9548	0.8952	0.9203	0.8158	0.8860
Effort [91]	0.9312	0.7975	0.9276	0.8536	0.8617
ForAda [17]	0.9544	0.7494	0.9395	0.8726	**0.9289**
FSFM-Adapter [86]	0.8689	0.8381	0.7516	0.8318	0.8486
Reference (PCR6$^+$-R)	0.9583	**0.9639**	0.9307	0.8530	0.8597

Method	DFEval24	FFIW	KoDF	IDForge	DFDM	Mean
DISCERN (local checkpoint)	0.6869	0.9036	0.8198	0.9723	0.9929	0.8959
GenD-CLIP [98]	0.6831	0.9278	0.8329	0.9839	**0.9959**	0.8945
DiCoME [48]	0.6769	0.9284	0.8509	0.9598	0.9909	0.8937
FSFM [86]	0.6817	0.9168	0.8467	0.9566	0.9790	0.8924
GenD-DINO [98]	0.6536	**0.9450**	0.8984	**0.9860**	0.9903	0.8883
GenD-PE [98]	0.6563	0.9330	0.8189	0.9808	0.9815	0.8842
Effort [91]	0.6769	0.9252	0.8725	0.9583	0.9729	0.8777
ForAda [17]	0.6227	0.9055	0.8739	0.9101	0.9713	0.8728
FSFM-Adapter [86]	0.6280	0.8565	0.7405	0.9077	0.8452	0.8117
Reference (PCR6$^+$-R)	**0.7235**	0.9217	**0.9600**	0.9631	0.9954	**0.9129**

subsets. The video view alone obtains 0.8857 and 0.9019 on these cells. Reenactment reverses this relationship: the spatial pair achieves 0.7481 and 0.8353, compared with 0.9355 and 0.9550 for the video view. The full framework combines these different strengths, achieving 0.9713 and 0.9834 on swapping and 0.9245 and 0.9616 on reenactment. Its family-balanced mean is 0.9602, and its generator-macro mean is 0.9588.

Figure 5.3 resolves this pattern at the generator level. For Celeb-DF-sourced DaNet, the appearance and latent facial-structure views achieve 0.4599 and 0.5289, while the video view reaches 0.9522. For TPSM, the corresponding scores are 0.5080, 0.5191, and 0.9915. The weak spatial scores in these cases contrast with high video discrimination. This is direct evidence that the video representation supplies useful information beyond the two spatial outputs, without requiring an assumption that spatial artifacts are absent from the underlying videos.

The pattern also occurs within cross-dataset benchmarks. On the reenactment subset

Table 5.3: Preliminary reference DF40 AUROC by manipulation family and source domain. A, L, and T denote appearance, latent facial-structure, and video evidence. Subsets use the same PCR6$^+$ coefficient-estimation procedure. The final row weights the four family–source cells equally. Bold denotes the best value in each row.

Family / source	T	$A+L$	$T+L$	$A+L+T$
Swap / CDF	0.8857	0.9674	0.9707	**0.9713**
Swap / FF++	0.9019	0.9709	0.9819	**0.9834**
Reenact. / CDF	**0.9355**	0.7481	0.9337	0.9245
Reenact. / FF++	0.9550	0.8353	**0.9633**	0.9616
Balanced mean	0.9195	0.8804	**0.9624**	0.9602

Table 5.4: Change in AUROC after removing one representation and re-estimating the remaining coefficients. A negative value indicates lower performance than the full reference. Benchmark groups are defined in the text; DF40 family means here average generator–source entries.

Evaluation group	Full	$-A$	$-L$	$-T$
Swap-dominated sets	0.9267	−0.0015	−0.0152	+0.0097
Reenactment-heavy sets	0.9620	+0.0027	−0.0092	−0.0941
In-the-wild sets	0.8226	−0.0024	−0.0083	−0.0201
DF40 face swap	0.9770	−0.0010	−0.0053	−0.0080
DF40 reenactment	0.9431	+0.0054	−0.0110	−0.1514

of CDF-v3, the spatial pair obtains 0.8430 and the full framework 0.9681. On KoDF's first-order-motion subset, the corresponding scores are 0.7310 and 0.9833. The video view alone is strong on both subsets, at 0.9741 and 0.9887. These results explain why the largest cross-dataset improvements arise on benchmarks containing substantial reenactment content.

Against the locally evaluated DiCoME comparator, the full system improves on 39 of 41 DF40 entries, including all 22 reenactment entries. The generator-macro margins are 0.1639 on reenactment and 0.0088 on swapping. The view-subset comparisons provide the more direct attribution: they show the contribution of video evidence while retaining the same integration procedure.

5.5.3 Contribution of Each Representation

Table 5.4 reports view-removal effects after re-estimating the remaining coefficients. For the cross-dataset analysis, the swap-dominated group contains CDF-v2, DFD, DFDC, DFDCP, IDForge, and DFDM; the reenactment-heavy group contains CDF-v3 and KoDF; and the in-the-wild group contains DeepfakeEval-2024 and FFIW. DF40 rows average generator–source

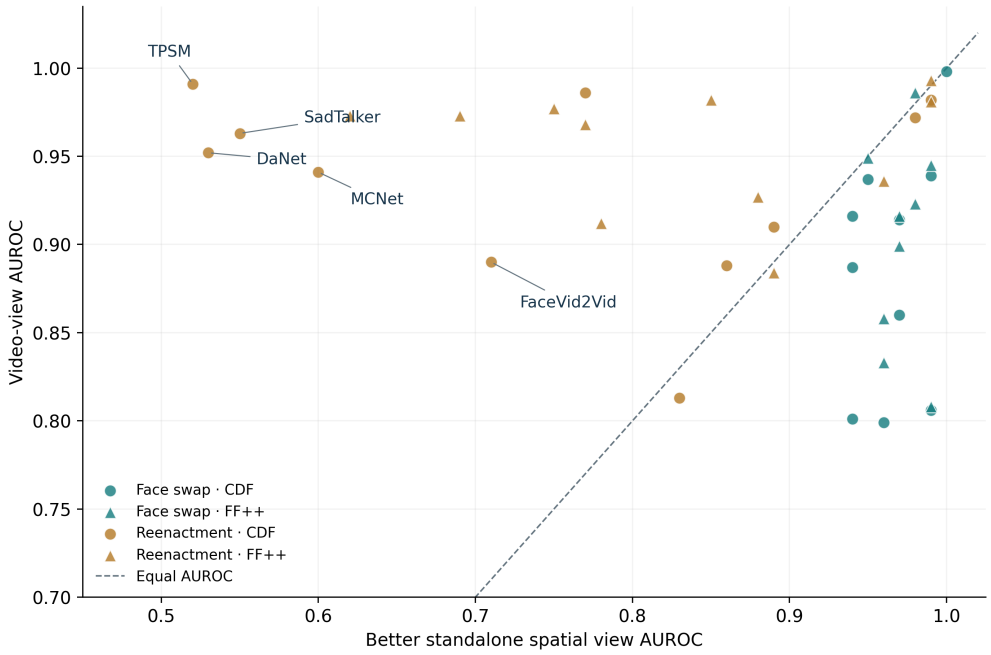


Figure 5.3: Representation complementarity over the 41 DF40 generator–source entries. The video view is compared with the better standalone spatial view on each entry. Reenactment includes several entries with weak spatial discrimination and high video-view AUROC. Points above the diagonal favor the video representation. Markers distinguish manipulation families and source domains; selected low-spatial-AUROC generators are annotated. Coordinates use the rounded per-generator table, and no confidence band is shown.

entries within each family.

The video representation has its largest contribution on reenactment. Removing it decreases AUROC by 0.0941 on the reenactment-heavy benchmarks and by 0.1514 on DF40 reenactment. It also contributes to the in-the-wild group and DF40 face swapping. On the swap-dominated cross-dataset group, the spatial pair instead has a 0.0097 advantage. This variation is consistent with the family-level complementarity observed in Table 5.3.

The latent facial-structure view contributes across all five groups: its removal lowers AUROC by 0.0053–0.0152. This broad empirical role complements the more pronounced family dependence of the video branch. The appearance view provides smaller gains on swap-dominated benchmarks, in-the-wild material, and DF40 swapping, while its removal slightly improves the two reenactment groups. Accordingly, the full system and the latent facial-structure–video pair are close in aggregate: their ten-benchmark means are 0.9129 and 0.9121, respectively, whereas the pair has the higher DF40 family-balanced mean of 0.9624.

Table 5.5 summarizes the independently refitted view subsets using the ten-dataset macro and four-cell DF40 mean. The $A + L$ ten-set mean and the separate local DISCERN row

Table 5.5: Preliminary reference view-subset comparison. Each subset has independently estimated reliability coefficients under the same protocol. Ten-set is the equal-dataset macro over the ten benchmarks in Table 5.2; DF40 weights its four family–source cells equally. Development-selection summaries are excluded from this evaluation table.

Views	Ten-set macro	DF40 family mean
T	0.8213	0.9195
$A{+}L$	0.8959	0.8804
$T{+}A$	0.9003	0.9522
$T{+}L$	0.9121	**0.9624**
$T{+}A{+}L$ (reference)	**0.9129**	0.9602

both round to 0.8959; they denote distinct models and are retained as separately reported aggregate results, not interchangeable checkpoints. The reliability coefficients reflect these differences in contribution. Across the five estimation repetitions of the primary configuration, the latent facial-structure coefficient is 1 and the video coefficient is 0.25; the appearance coefficient is 0.25 in two repetitions and zero in three. These are fixed model-level choices. They show that the formulation can retain a three-view representation while reducing or suppressing a view when its additional evidence is less useful under the selection objective.

The fusion comparison does not establish a detection advantage for PCR6$^+$. The ten-dataset means are 0.9129 for direct PCR6$^+$, 0.9134 for direct PCR6, 0.9147 for pairwise PCR6, and 0.9141 for opinion averaging. These differences are small relative to the video-removal effects on reenactment. The remaining investigation therefore keeps fusion fixed and tests representations and uncertainty training.

5.5.4 Uncertainty and Selective Prediction

Table 5.6 evaluates three uncertainty scores on the same predictions: fused vacuity, retained conflict, and predictive entropy. Vacuity gives the strongest error ranking on CDF-v3, DFD, DeepfakeEval-2024, and KoDF among the seven benchmarks discussed here. Entropy is stronger on CDF-v2, DFDC, and FFIW. The complete ten-benchmark table reports the corresponding macro means, computed from the same per-benchmark records. Evidence strength therefore supplies useful ranking information, with the relative benefit depending on the benchmark.

Figure 5.4 illustrates the resulting selective-prediction behavior at two measured coverage levels. On CDF-v3, reducing coverage from 90% to 80% decreases the retained error rate from 17.41% to 14.54%; on DFD, it decreases from 16.92% to 13.94%. KoDF has lower retained risks of 5.97% and 5.22%, respectively. All seven plotted benchmarks show a reduction

Table 5.6: Uncertainty assessment for the preliminary reference. Error-detection AUROC (first three columns) compares scores for the same predictions and fixed decision threshold; higher is better. ECE, NLL, and excess area under the risk–coverage curve (E-AURC, using vacuity) are lower-is-better. Bold identifies the best error-ranking score per benchmark.

Benchmark	Vacuity ↑	Conflict ↑	Entropy ↑	ECE ↓	NLL ↓	E-AURC ↓
CDF-v2	0.641	0.665	**0.667**	0.254	0.516	0.1119
CDF-v3	**0.768**	0.542	0.642	0.283	0.517	0.0547
DFD	**0.790**	0.715	0.750	0.220	0.456	0.0470
DFDC	0.672	0.667	**0.708**	0.105	0.530	0.0983
DFDCP	**0.675**	0.643	0.668	0.180	0.570	0.1046
DFEval24	**0.627**	0.568	0.614	0.188	0.760	0.1855
FFIW	0.699	0.754	**0.785**	0.099	0.405	0.0749
KoDF	**0.821**	0.655	0.718	0.128	0.345	0.0199
IDForge	0.793	0.782	**0.846**	0.078	0.285	0.0314
DFDM	0.897	0.870	**0.908**	0.060	0.108	0.0034
Macro mean	0.738	0.686	0.731	0.160	0.449	0.0732

between these two operating points. This supports using vacuity to prioritize uncertain videos for further review.

The preliminary reference has a substantial probability-calibration weakness. Its ECE is 0.254 on CDF-v2 and 0.283 on CDF-v3, compared with DISCERN's published 0.014 and 0.027, respectively. These correspond to approximately 18-fold and 10-fold differences, although different checkpoints, evaluation pipelines, and 15 versus 10 bins prevent interpreting the ratios as controlled effects. Good AUROC or error ranking does not resolve this weakness. A validation-fitted temperature-scaling comparator is therefore required in the matched uncertainty study.

Error ranking and probability calibration assess different properties. The ECE and NLL columns in Table 5.6 characterize the latter separately, and calibration varies considerably across datasets. Thus the selective-prediction result concerns the ordering of decisions by uncertainty; it does not require interpreting the numerical probability as a universally calibrated deployment confidence.

Conflict has a complementary diagnostic role. In the development analysis, video–spatial disagreement is high both when the video view is correct and the spatial views are wrong, and when the opposite occurs. Conflict identifies incompatible support but does not by itself determine which representation should be trusted. Reporting it alongside vacuity preserves this distinction.

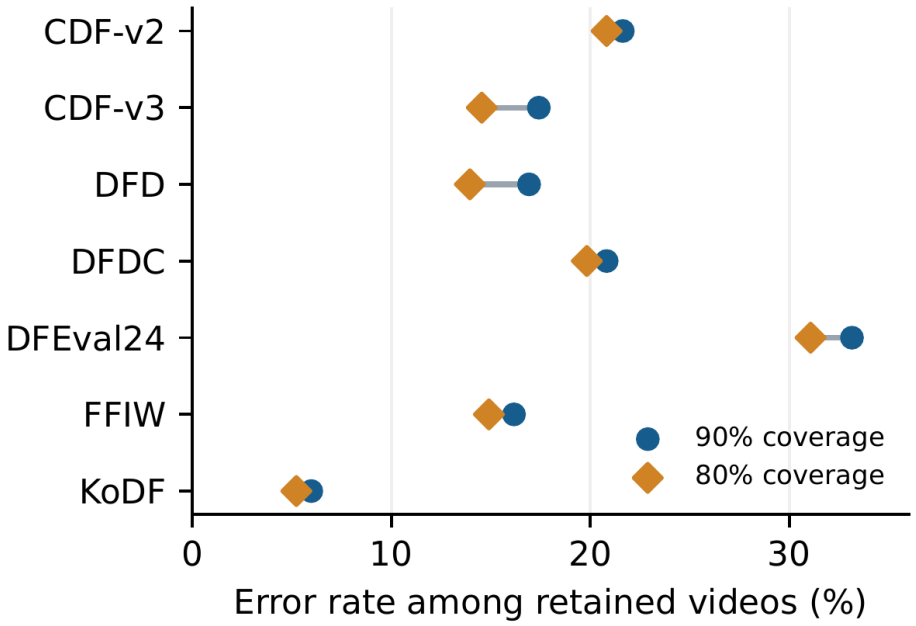


Figure 5.4: Selective prediction using fused vacuity. At each coverage level, videos with the lowest vacuity are retained. Reducing coverage from 90% to 80% lowers the observed error rate on all seven plotted benchmarks. Points are measured operating summaries, rather than interpolated risk–coverage curves.

5.5.5 Image Degradation, Configuration Variation, and Adaptation Cost

The reference is evaluated under JPEG compression and resampling in addition to the clean crop exports. On CDF-v2, AUROC changes from 0.9583 to 0.8683 under JPEG quality 40 and to 0.9396 under resampling at scale 0.60. On DFDCP, the corresponding scores are 0.8597, 0.8170, and 0.8033. The degradation depends on both the benchmark and the transformation, with compression producing the larger reduction on CDF-v2. These measurements characterize robustness to different severities of familiar augmentation types; the associated uncertainty measurements are retained in the experimental record.

The four detector configurations achieve DF40 family-balanced AUROCs of 0.9602, 0.9651, 0.9635, and 0.9690. The first two differ in video-branch training seed while sharing the spatial checkpoint; the other two use an alternative spatial training recipe with matched spatial and video seeds. This analysis examines both video-branch initialization and a change in the underlying spatial system. It complements the five coefficient-estimation repetitions, which do not retrain the encoders.

The reference video extension, with its 128-unit head, adds 786,432 trainable adapter

parameters and 133,506 head parameters, totaling 919,938. Three reliability coefficients are estimated separately. This is a parameter-efficient adaptation of a pretrained video encoder; the full inference system still executes the spatial and video backbones. The count describes additional trainable parameters in the reference. CROSS-EXAM's 256-unit head and separate PE-Core encoder require their own parameter and runtime measurements.

5.6 Evaluation of CROSS-EXAM

5.6.1 Development, Selection, and Difficult-Dataset Evaluation

The prospective experiments train on FF++ c23 and preserve source grouping when constructing partitions. Genuine videos, derived manipulations, and processing variants from one source remain together. Feature normalization and nuisance bases use the training portion only. Development data select suppression rank, regularization strengths, checkpoints, and reliability coefficients. Source-grouped outer assessment folds are excluded from each fitting and selection step. DeepfakeEval-2024 and DFDC evaluation source groups are reserved before these new comparisons and excluded from training and selection; the remaining development pool selects configurations by overall macro AUROC.

CDF-v2 and DeepfakeEval-2024 informed the reference's development, and DF40 informed earlier fusion choices. These benchmarks remain valuable for continuity and family analysis, but they are development-exposed. DeepfakeEval-2024 and DFDC are the primary and secondary difficult-dataset stress tests, respectively. Their prior exposure remains explicit; a new source split does not erase earlier benchmark feedback. Independent training seeds assess optimization variability; they do not remove bias from benchmark reuse.

The primary architecture uses PE-Core at 448×448, layer-norm adaptation, and nuisance suppression. Stage 1 uses separately supervised branches. Stage 2 compares no IBDC, original symmetric IBDC, and one-sided IBDC using the same three trainable heads. The no-IBDC control receives the same Stage 2 EDL updates. This separates the effect of the new regularizer from the effect of extra head training.

5.6.2 Representation and Loss Comparisons

The representation experiment compares the shared-CLIP reference, a separate PE-Core appearance path without suppression, and the complete APP path with suppression. Each is evaluated against the LFS–TD pair. The first comparison holds the LFS and TD checkpoints fixed to isolate the APP change. Subsequent complete adaptation repeats measure the

Table 5.7: Primary CROSS-EXAM comparisons. Selection partitions and threshold rules are shared within each matched contrast.

Question	Matched contrast	Main evidence
Appearance contribution	Separate PE-Core versus shared CLIP; each full system versus LFS–TD	Dataset/family AUROC; refitted removal effect; error overlap
Nuisance suppression	No suppression, selected directions, partial attenuation, matched random directions	Clean/processed detection; genuine-video FPR; incremental APP value
Uncertainty learning	One-sided versus original and no IBDC; heads only in Stage 2	E-AURC; error AUROC; ECE; NLL; detection difference
Post-hoc calibration	No-IBDC model plus one temperature fitted on FF++ validation; compare calibrated versions of all loss arms	NLL; Brier; ECE; AUROC unchanged by monotone scaling
Evidence regimes	Weak agreement, confident agreement, committed disagreement	Vacuity response; shared errors; correct specialist behavior
Video information	Ordered, shuffled, repeated-frame, trained order-invariant control	Paired detection difference and family effects

selected system's variability. Each view subset has reliability coefficients fitted by the same permitted procedure before its Stage 2 comparison.

The suppression experiment matches training images, paired transformations, optimization steps, head capacity, and selection data. It compares ordinary augmentation alone, the selected nuisance projection, partial attenuation, and a random projection of equal rank. Clean detection, processed-video detection, genuine-video false alarms, and APP's refitted removal effect determine whether suppression removes nuisance variation while preserving useful forensic evidence. A frozen PE-Core control additionally isolates the contribution of layer-norm adaptation.

The loss experiment crosses appearance configuration with no, original, and one-sided IBDC. Within each loss comparison, encoders, nuisance operators, coefficients, head initialization, and trainable parameter sets are identical. This comparison asks whether one-sided IBDC improves uncertainty for fixed features, and whether its effect depends on the representation set. Fusion remains direct multisource PCR6^{+} throughout the primary comparisons; the preliminary fusion results do not support claiming a detection gain from this rule alone.

5.6.3 Detection, Uncertainty, and Operating Behavior

Video-level AUROC is reported by dataset and manipulation family, followed by the prespecified macro mean. The reference's ten-dataset mean, four-cell DF40 family mean, and 41-entry generator mean retain their distinct definitions. A threshold fitted at the FF++ validation equal-error point is transferred unchanged to each evaluation set. Fixed-threshold FPR and TPR accompany AUROC, including for genuine videos subjected to the same processing transformations as manipulated videos.

The primary uncertainty outcome is E-AURC using fused vacuity. Lower values indicate an error ordering closer to the optimal ordering at the model's own error rate. Complete risk–coverage curves, full-coverage error, and retained risk at 90% and 80% coverage accompany this summary. Error-detection AUROC compares vacuity, predictive entropy, and total conflict on the same predictions. Probability quality is evaluated separately using NLL, Brier score, and ECE with 15 equal-width bins. Neither good error ranking nor low ECE alone establishes reliable decisions across target domains.

Shared ignorance is identified from low evidence in all views and close class probabilities, with cutoffs fixed using development predictions. Confident agreement and committed disagreement form complementary strata. Labels are used to evaluate error behavior within these strata, not to define the strata at inference. Particular attention is given to correct specialist predictions: raising uncertainty whenever a specialist disagrees could weaken the representation complementarity that motivates CROSS-EXAM.

The development screen permits a macro-AUROC reduction of at most 0.005 relative to the matched control when accompanied by improved E-AURC or processing robustness. A formal noninferiority claim requires the lower confidence bound of the paired AUROC difference to exceed -0.005; the point-estimate screen alone is insufficient. Dataset-specific results and fixed-threshold errors are reported even when the macro comparison is favorable.

Temperature-scaling control. For a fused probability p_f, form $z = \log[p_f/(1-p_f)]$ after numerical clipping and return $p_T = \sigma(z/T)$, with one scalar $T > 0$ fitted by minimizing NLL on FF++ validation only [35]. The temperature is frozen for all target datasets. This monotone transformation preserves AUROC, but can change NLL, ECE, and fixed-threshold decisions. The raw probability and vacuity remain available: temperature scaling does not redefine Dirichlet strength. Each no-, original-, and one-sided-IBDC arm is evaluated before and after the same calibration procedure. An IBDC calibration claim must therefore exceed the gain available from this one-parameter control.

5.6.4 Video Controls and Statistical Analysis

The input-order test permutes frames before V-JEPA 2 and recomputes its features. A shuffle of cached output tokens cannot substitute for this intervention because the encoder has already processed the original sequence. A repeated-frame clip removes between-frame variation. A separately trained order-invariant model averages frame features using a matched evidence-head capacity. These controls distinguish order-sensitive information from the benefit of observing multiple facial states.

Paired source-grouped bootstrap resampling uses the same evaluated videos for each contrast. Confidence intervals characterize sample uncertainty conditional on the trained models. Three complete independent adaptation runs assess training variability for the retained comparison; they are distinct from the reference's repeated coefficient fits. Results are summarized both per seed and across seeds, preserving the pairing between each model and its corresponding control.

5.6.5 Reproducibility and Computational Cost

Each experiment records encoder and adapter checkpoints, feature layer, preprocessing, source partitions, seed, nuisance rank and basis, head dimensions, optimizer schedule, and fusion coefficients. Per-video exports contain labels, branch evidence, pooled opinions, prediction, vacuity, and conflict. These exports support consistent tables and risk–coverage curves without reconstructing metrics from rounded summaries.

Processing coverage is reported by dataset and class, including face-extraction failures and short clips. Computational measurements include facial preprocessing, all executed backbones, head computation, and fusion. Trainable parameters describe adaptation efficiency; total parameters, memory, latency, and throughput describe execution cost. PE-Core's 448×448 input is therefore part of the cost comparison as well as the representation specification.

5.7 Limitations and Expected Contribution

Separate encoders do not guarantee complementary errors. They can learn correlated shortcuts, share pretraining content, or rely on the same defective face crops. Nuisance projection is similarly selective: a direction associated with compression may also carry manipulation information. The matched controls and family-resolved analysis determine whether the combined system benefits from the intended mechanisms.

One-sided IBDC addresses committed disagreement and the certainty incentive associated with weak agreement. It does not detect all shared confident errors, and its use of product

vacuity can reduce useful specialist evidence. These are empirical limits of the training design, assessed jointly with detection rather than inferred from the loss definition.

The expected contribution is a method for cross-examining separately learned facial evidence: processing-aware appearance, latent facial structure, and video representations are trained individually, then their evidential heads are coordinated under a one-sided conflict constraint. The preliminary reference establishes spatial–video complementarity; the prospective comparisons test the incremental value of the complete CROSS-EXAM design. A supported conclusion may favor a simpler subset or the no-IBDC control, while still resolving the research questions about representation contribution and uncertainty learning.

CHAPTER 6

Research Plan and Expected Contributions

The dissertation progresses from facial-signature learning in DBaGNet, to heterogeneous evidence and uncertainty in DISCERN, and then to CROSS-EXAM's separate representations and evidence-sensitive head training. The completed studies establish the foundation for RQ1 and RQ2. The research plan for RQ3 tests the appearance path, nuisance suppression, and one-sided IBDC through matched comparisons, followed by difficult-dataset evaluation and dissertation integration.

6.1 Completed Research and Empirical Foundation

DBaGNet evaluates whether identity, behavioral, and geometric signatures support transfer across manipulation methods and datasets. Its feature ablations provide the principal evidence for combining those signatures. DISCERN extends the investigation to visual, semantic, and structural evidence, with detection, calibration, selective-prediction, and corruption experiments. Its final input specification comprises 50 semantic features, 82 forensic features, and 18 selected predicates.

The preliminary third-study reference establishes complementary spatial and video behavior across ten benchmarks and the evaluated DF40 generator–source entries. The video view contributes strongly on reenactment, while latent facial structure contributes across the reported groups. Appearance adds relatively little when it shares the CLIP feature path. These observations motivate CROSS-EXAM's separate PE-Core appearance encoder and the LFS–TD pair as a required comparison. DISCERN's shared-ignorance behavior motivates the one-sided loss.

6.2 Work Packages and Deliverables

6.2.1 Experiment Specification and Reference Reproduction

The first package associates each reference result with its encoder and adapter checkpoints, crop exports, source partitions, reliability coefficients, and per-video predictions. The locally evaluated DISCERN comparator is kept distinct from the accepted-paper protocol. The deliverable is a reproducible reference against which the CROSS-EXAM contrasts can be interpreted.

CROSS-EXAM uses PE-Core at 448×448 with layer-norm adaptation and nuisance suppression, CLIP with LoRA and MR-VAE for LFS, and V-JEPA 2 with LoRA for TD. Stage 1 updates each branch separately with EDL and the relevant auxiliary objectives. Stage 2 updates the three evidential heads only. These specifications define the system studied by the following work packages.

6.2.2 Representation Complementarity and Processing Robustness

The second package compares the separate PE-Core branch with shared-CLIP appearance and the LFS–TD pair. It then isolates nuisance suppression through ordinary augmentation, partial attenuation, and rank-matched random-direction controls. All comparisons preserve source grouping and matched training exposure.

The deliverable combines standalone detection with the incremental effect of APP on the assembled detector. Dataset- and family-level results, refitted branch removal, joint error patterns, and processing-induced false alarms determine whether the appearance view adds useful evidence. Computational cost accompanies each comparison because the separate 448×448 encoder changes inference requirements.

6.2.3 Joint Evidential-Head Training

The third package compares original, one-sided, and no IBDC, together with validation-fitted temperature scaling, using identical fixed representations and identical trainable head modules. Each control receives the same Stage 2 supervised EDL training. E-AURC is the primary uncertainty outcome, supported by risk–coverage curves, error-detection AUROC, ECE, NLL, and Brier score.

The deliverable explains how each objective behaves during shared ignorance, confident agreement, and committed disagreement. Detection and branch-removal effects determine whether improved error ranking comes at the cost of useful specialist evidence. The mathe-

matical removal of a certainty incentive and the empirical value of the new loss are evaluated as separate claims.

6.2.4 Difficult-Dataset Evaluation and Practical Characterization

The fourth package evaluates the retained configuration using contrasts fixed during development. DeepfakeEval-2024 is the primary difficult-dataset stress test and DFDC is the secondary stress test. Their reserved evaluation source groups are excluded from new representation selection, nuisance-basis fitting, head training, reliability fitting, and threshold selection. Both datasets have prior development exposure, so the results assess known difficult conditions rather than untouched-data confirmation. Three complete adaptation runs assess training variability. Paired source-grouped resampling assesses uncertainty in the comparative results.

The deliverable includes fixed-threshold false-positive and true-positive rates, processing coverage, compression and resampling behavior, video-order controls, and complete inference cost. A confidence interval that spans the decision boundary is reported as inconclusive. Benchmark results obtained after development exposure remain identified separately from confirmation.

6.2.5 Manuscript and Dissertation Integration

The final package connects the three studies through their research questions and consolidates the method, results, limitations, and reproducibility record. The journal target for CROSS-EXAM is IEEE Transactions on Biometrics, Behavior, and Identity Science. The planned deliverables are an advisor-reviewed journal submission and a dissertation that explains both the gains and the limits of the selected evidence design.

6.3 Research Schedule and Milestones

Table 6.1 and Figure 6.1 organize the remaining work around explicit decision points. November 3, 2026 is the planning target for the proposal defense, with circulation by October 20 to provide two weeks for review. March 2027 is the planning window for the final defense. These are scheduling targets, subject to advisor and committee availability. The experimental sequence first fixes the appearance representation, then nuisance suppression, and finally the head-training objective; an earlier decision's fallback becomes the input to the next comparison.

Table 6.1: Research schedule and planned milestones. Representation, suppression, and uncertainty decisions are sequential; manuscript drafting can proceed alongside evaluation.

Window	Activity	Milestone or deliverable
October 12–23, 2026	Reference audit, proposal revision, and advisor review	Circulate by October 20 for the planned November 3 proposal defense
November 3, 2026	Proposal defense	Committee feedback and approved research scope
November 4–20, 2026	Matched appearance and view-subset comparison	D1: retain APP or select LFS–TD
November 23–December 11, 2026	Processing-nuisance comparison, conditional on D1	D2: retain suppression or ordinary augmentation; skip if APP is absent
December 14, 2026–January 15, 2027	Matched head-only IBDC and temperature-scaling comparisons	D3: retain one-sided IBDC or matched no-IBDC control
January 18–February 12, 2027	Independent adaptation repeats, difficult-dataset evaluation, and cost analysis	Complete matched results and reproducibility package
February 2027	Journal manuscript and dissertation integration	Target T-BIOM submission by February 19
March 2027	Committee review and dissertation defense preparation	Target final defense in late March; committee-coordinated scheduling

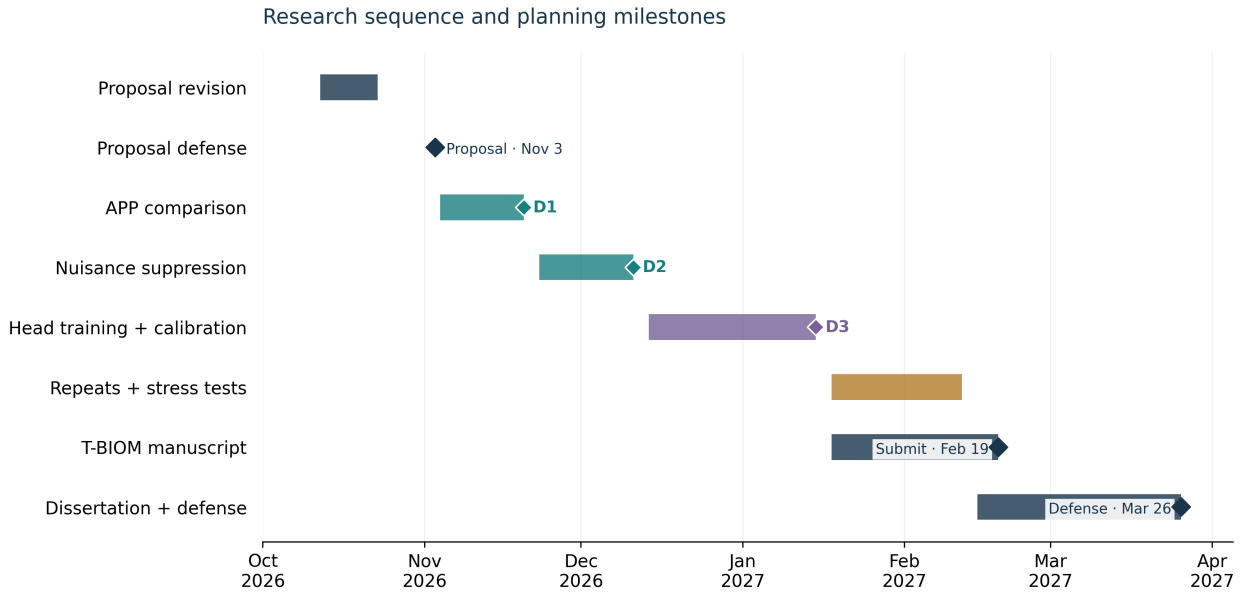


Figure 6.1: Planned research timeline. Bars denote work periods, colored diamonds mark decisions D1–D3, and dark milestone markers identify proposal defense, T-BIOM submission, and final defense targets. Experimental dependencies follow D1, then D2, then D3. Calendar targets do not imply confirmed committee appointments.

6.4 Completion Criteria and Contingencies

Figure **??** shows how each outcome yields a complete detector and an answer to the corresponding research question.

6.4.1 Representation Decision

APP is retained when a matched comparison establishes an incremental detection, uncertainty, or processing-robustness benefit over the strongest simpler system at an acceptable computational cost. A stronger standalone branch or a nonzero reliability coefficient is insufficient by itself. The comparison re-estimates coefficients for each view subset and evaluates that assembled system.

If the separate appearance path remains redundant, the LFS–TD pair is a valid outcome. If suppression removes useful forensic information, the same PE-Core branch without suppression provides the matched alternative. These outcomes answer the representation question without requiring an unrestricted backbone search.

6.4.2 Uncertainty Decision

One-sided IBDC is retained if it improves the prespecified uncertainty outcome and meets the detection criterion in Chapter 5. The decision considers high-conflict correct predictions and individual datasets as well as the macro result. Calibration, error ranking, and selective prediction retain their separate interpretations.

If the loss weakens detection or specialist contributions without the required uncertainty benefit, the matched no-IBDC system supplies the alternative. The dissertation can establish the shared-ignorance distinction mathematically while concluding that the evaluated regularizer does not improve the selected operating criteria. Such a result narrows the method's supported claims rather than leaving the research question unanswered.

6.4.3 Data and Compute Priorities

The primary representation, suppression, and loss contrasts take priority, followed by independent adaptation repeats and difficult-dataset evaluation. Additional diagnostics are selected to resolve a specific remaining question. If a previously untouched evaluation group cannot be reserved, the analysis reports development-exposed benchmark evaluation rather than claiming untouched confirmation. If source identifiers are incomplete, resampling uses the grouping that can be established and states its scope.

6.5 Expected Contributions

The dissertation is expected to establish three connected contributions. DBaGNet shows how meaningful facial signatures can support transfer under its reported protocols. DISCERN connects heterogeneous evidence to detection and uncertainty assessment while exposing conditions under which disagreement is insufficient. CROSS-EXAM provides a controlled method for testing separate facial representations, processing-nuisance suppression, and joint evidence-head training.

Together, the studies seek a more precise account of generalizable detection: which observations contribute beyond existing views, which processing changes alter that contribution, and whether the resulting uncertainty helps identify unreliable decisions. The final conclusions will follow the matched evidence for those questions rather than the number of branches or the nominal sophistication of the fusion rule.